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TIME SERIES FORECATING USING 91-DAY TREASURY BILL RATE

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DECLARATION

This project is submitted as part of fulfilment for the award of a **BTech** in Statistics The work is a result of my investigation. All section of the text and results which have been obtained from other works/ sources are fully referenced. I understand that cheating and plagiarism constitute a breach of Accra Technical University and will be dealt with accordingly.

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DECLARATION BY SUPERVISOR

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DEDICATION

This book is dedicated to the Almighty God, my parents, my mentor, my siblings, my friends, and all of my professors for their unwavering support and guidance during my training.

ACKNOWLEDGEMENTS

I want to use this chance to express my appreciation to everyone who helped make this endeavor a success. But without the generous assistance and support of my classmates, it would not have been feasible. I want to express my gratitude to each and every one of them. I owe a debt of gratitude to my mentor, supervisor, and head of department for their unwavering supervision, direction, and provision of the knowledge I needed to finish the job. I would want to thank my mother for her helpful assistance and support in seeing this project through to completion.

ABSTRACT

This study presents a comprehensive analysis of time series forecasting models applied to Ghana's 91-day Treasury Bill rates, utilizing multiple methodological approaches to develop robust predictive frameworks for this critical monetary policy instrument. The research employed 381 daily observations spanning from June 2017 to May 2025, capturing a period of significant monetary volatility and structural changes in Ghana's financial environment.

The analysis integrated traditional econometric methods with advanced machine learning techniques to forecast Treasury Bill rates. Traditional approaches included ARIMA models, SARIMA models, and Holt-Winters multiplicative methods, while machine learning implementations encompassed Artificial Neural Networks (ANN), Facebook Prophet, and Long Short-Term Memory (LSTM) models. The study addressed the non-stationary nature of the original time series through first-order differencing, with stationarity confirmed via Augmented Dickey-Fuller tests.

Key findings revealed substantial variability in Treasury Bill rates over the study period, with a mean rate of 18.98% and standard deviation of 6.97%. The data exhibited distinct phases: relative stability between 2017 and mid-2022 (rates maintained between 12-15%), followed by unprecedented volatility in 2022-2023 with peaks reaching approximately 36%, before stabilizing around 15% in 2025. This structural break presented significant challenges for all forecasting methodologies.

Among the models evaluated, the ARIMA (1,1,0) model demonstrated superior performance with the lowest AIC value of 761.95, effectively capturing the autoregressive characteristics while maintaining parsimony. The Facebook Prophet model successfully identified structural breaks and provided appropriate confidence intervals reflecting increased uncertainty. Machine learning approaches showed mixed results: ANN exhibited strong generalization capabilities, while LSTM demonstrated overfitting tendencies on test data.

Forecasting projections from all methodologies converged on a stabilization of policy rates within the 15-20% range by 2026, indicating a return to normalized monetary conditions following the 2022-2023 volatility period. The study concluded that traditional econometric methods, particularly ARIMA models, remain competitive with advanced machine learning approaches for Treasury Bill rate forecasting, offering superior interpretability and robustness for policy applications.

The research provides practical implications for multiple stakeholders: investors can utilize these forecasts for portfolio optimization and risk management, policymakers can better understand evolving monetary conditions, and financial institutions can enhance asset-liability management strategies. The study contributes to the emerging markets interest rate forecasting literature while demonstrating the continued relevance of traditional time series methods in contemporary monetary analysis.

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CHAPTER ONE

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

Treasury bills are the largest financial instruments in Ghana's money market and play a critical role in the country's monetary policy framework. They are short-term debt securities issued by the Government of Ghana through the Bank of Ghana and serve as a benchmark for interest rates across the economy while providing a mechanism for government borrowing and liquidity management (Quartey & Afful-Mensah, 2014). Treasury bills have progressed since their introduction in Ghana's financial system. Treasury bills have developed gradually to become essential tools for monetary policy implementation, fiscal deficit financing, and investment opportunities for both institutional and individual investors (Osei-Assibey, 2016) Ghana's treasury bond market has undergone significant changes over the past three decades, reflecting the country's macroeconomic transformation and the development of the financial sector. Following financial sector reforms initiated under the Economic Recovery Program in the 1980s, Ghana's securities market gradually developed and diversified (Bawumia et al., 2008). With the adoption of an inflation-targeting regime by the Bank of Ghana in 2002, the importance of treasury bill rates as a monetary policy instrument increased further, creating a direct link between policy decisions and market interest rates (Alagidede and Ibrahim, 2017).

However, the Ghanaian Treasury bill market has been highly volatile, with interest rates fluctuating due to various domestic and global economic factors. Inflationary pressures, fiscal imbalances, exchange rate depreciation, and global financial conditions have contributed to the dynamic behavior of Treasury bill interest rates in Ghana (Aryeetey and Harrigan, 2000). For example, from 2000 to 2022, the 91-day Treasury bill rate ranged from 9.5% to 27.8%, indicating a high degree of volatility that poses significant challenges for investors, policymakers, and financial institutions (BoG, 2022). As noted by Boamah (2017), accurately predicting Treasury bill rates is very important for various stakeholders in the Ghanaian economy. For economic policymakers, having reliable forecasts can help them make decisions on monetary policy and debt management strategies. Financial institutions rely on interest rate forecasts to manage assets and liabilities and price financial products. Investors, both domestic and foreign, rely on interest rate forecasts when making decisions about portfolio allocation and risk management (Agyei-Ampomah, 2011).

1.2 STATEMENT OF THE PROBLEM

Despite the critical importance of treasury bill interest rates in the Ghanaian financial system, forecasting these rates remains challenging due to several factors. As (Addison, 2018) noted, treasury bill interest rates in Ghana are influenced by a complex interplay of fiscal, monetary, and external factors, making their forecasting difficult using conventional approaches. Significant government borrowing needs, patchy inflation patterns, and volatile exchange rates create a complex environment for interest rate setting (Quartey, Ackah, & Agyei-Holmes, 2017). The existing literature on forecasting treasury bill interest rates in developing countries, particularly in sub-Saharan Africa, remains limited compared to the extensive research in advanced economies. As (Acquah-Sam & Salami, 2014) highlighted, most forecasting models have been developed and tested in the context of developed financial markets with different structural characteristics than those in emerging markets such as Ghana. This gap raises questions regarding the applicability and accuracy of conventional forecasting methods in the Ghanaian context. Conventional time series forecasts often fail to capture the nonlinear dynamics and structural discontinuities that characterize Ghanaian government bond interest rates (Anoki & Toyneboah, 2009). Economic shocks, political regime changes, and institutional reforms have disrupted historical interest rate trends and posed methodological challenges for forecasting (Frimpong & Oteng-Abayai, 2008). (Beckbe, 2011) also argues that the limited availability of high-quality, high-frequency data in Ghanaian financial markets limits the application of sophisticated forecasting techniques that would improve forecast accuracy. These data limitations, coupled with the unique institutional characteristics of the Ghanaian government bond market, require careful consideration of the most appropriate and effective forecasting methods.

1.3 RESEARCH QUESTIONS

Based on the identified problems, this study seeks to address the following research questions:

- What are the main patterns and characteristics in the time series behavior of Ghana's 91-day Treasury bill rates over the study period?
- Which time series forecasting models provide the most accurate forecasts of Ghana's 91-day Treasury bill rates over different forecasting models?
- How do traditional econometric forecasting methods, such as ARIMA, compare with machine learning methods in forecasting Treasury bill rates in Ghana?

- To what extent do seasonal patterns affect the behavior and predictability of Treasury bill rates in Ghana?
- How have structural changes in Ghana's economy and financial system affected the stability and predictability of Treasury bill rate trends?

1.4 OBJECTIVES OF THE STUDY

The primary objective of this study is to develop and evaluate time series forecasting models for predicting Treasury bill rates in Ghana. The specific objectives are:

- Analyze the historical behavior of Ghanaian 91-day Treasury bill interest rates and identify key patterns, including trend, seasonal, and cyclical components.
- Develop and compare alternative forecasting models, including ARIMA, SARIMA, exponential smoothing, and machine learning models for Treasury bill rate prediction.
- Evaluate the forecasting performance of different models using appropriate accuracy metrics and statistical tests.
- Assess the impact of seasonality and structural breaks on the predictability of Treasury bill rates in Ghana.
- Provide evidence-based recommendations for forecasting Ghanaian Treasury bill rates that inform the decision-making process of policymakers, financial institutions, and investors.

1.5 SIGNIFICANCE OF THE STUDY

This research contributes to theoretical knowledge and practical application in several important ways. From a theoretical perspective, the study fills an important gap in the literature by examining the applicability of different time series forecasting methods in the context of an emerging economy with unique financial market characteristics. As (Adam, 2014) points out, most empirical studies on interest rate forecasting have focused on developed markets, with little attention to the specific challenges of developing economies. The study will also contribute to the growing literature on the relative performance of traditional econometric methods versus machine learning approaches in financial forecasting. According to Siami-Namini et al. (2018), machine learning models have shown promising results in several forecasting applications, but their effectiveness for specific financial variables in emerging markets requires further research. This study provides empirical evidence on this important methodological issue in the Ghanaian context.

In practice, accurate forecasts of Treasury bond prices have a significant impact on the implementation of monetary policy in Ghana. As (Bomia & Abrado-Otto, 2003) point out, the effectiveness of monetary policy depends in part on the feasibility of interest rates and their transmission to other sectors of the economy. By developing a better forecasting model, this study can improve the Bank of Ghana's ability to make forward-looking monetary policy decisions. For financial institutions operating in Ghana, better forecasts of Treasury bond prices can improve asset and liability management, risk assessment, and pricing strategies. (Mensah, 2013) highlights how interest rate volatility poses significant challenges for banks and other financial intermediaries in Ghana. The forecasting methods developed in this study can help these institutions to anticipate market movements more accurately and manage their exposures accordingly. Investors in Ghana's financial markets, including pension funds, insurance companies and individual savers, rely on Treasury bond prices as a benchmark for evaluating investment opportunities and as a key component of their portfolios. As (Osei, 2002) notes, Treasury bonds are the most popular financial asset among Ghanaian investors. Improved forecasting methods can help these investors make more informed portfolio allocation and risk management decisions. Finally, the study has implications for fiscal policy and debt management in Ghana. According to (Oduro-Afriyie & Adeyemi, 2020), government borrowing costs, as reflected in Treasury bond prices, have a significant impact on fiscal sustainability and public debt dynamics. By improving the predictability of Treasury bond prices, this research could contribute to the effectiveness of the Ministry of Finance's debt management strategies and financial planning.

1.6 SCOPE AND LIMITATIONS

The study will focus on analyzing and forecasting the 91-day Treasury bill rates in Ghana, covering the period from January 2006 to December 2022. This timeframe encompasses various phases of Ghana's economic development, including periods of high growth, fiscal challenges, and external shocks, providing a comprehensive dataset for model development and testing. While the research aims to be thorough, several limitations are acknowledged. First, as noted by (Ayitey, 2019), the quality and consistency of financial data in Ghana have improved over time but still present challenges, particularly for earlier periods. Data limitations may affect the precision of certain analyses and the applicability of some advanced modeling techniques.

Also, the study focuses primarily on quantitative forecasting methods and may not fully capture qualitative factors, such as market sentiment and policy credibility, that can influence T-bill

rates. As (Biekpe & Adjasi, 2006) argue, market psychology and institutional factors play important roles in Ghana's financial markets but are difficult to incorporate into formal forecasting models. The research concentrates on forecasting T-bill rates rather than explaining the structural relationships between T-bill rates and other economic variables. While the study will examine the predictive value of including external factors in forecasting models, a comprehensive structural model of interest rate determination in Ghana lies beyond its scope.

As with all forecasting exercises, the models developed in this study are subject to the inherent unpredictability of financial markets and cannot account for unprecedented events or regime changes that might occur in the future. As (Mensah, Forecasting uncertainty in Ghanaian financial markets, 2015) cautions, all forecasting models should be viewed as tools for reducing uncertainty rather than eliminating it. Despite these limitations, the study represents a significant contribution to understanding and predicting Treasury bill rates in Ghana, with important implications for policy, investment, and financial management decisions.

1.7 ORGANIZATION OF THE STUDY

The study is organized as follows:

CHAPTER ONE: Present the Background, Statement of the Problem, Research Questions, Objectives of the Study, Significance of the Study, Scope and Limitations, and Organization of the Study.

CHAPTER TWO: Literature Review - This chapter provides a comprehensive review of relevant literature on treasury bill rate forecasting, organized by model type and geographical context. It also establishes the theoretical framework connecting macroeconomic theory to treasury bill rate behavior.

CHAPTER THREE: Methodology - This chapter details the research design, data sources, sampling techniques, model specification procedures, and evaluation metrics used in the study.

CHAPTER FOUR: Result and Discussion - This chapter presents the empirical findings, including model estimation results, comparative performance analysis, and forecast evaluations, along with visualizations and interpretations.

CHAPTER FIVE: Limitations, Conclusion, and Recommendation of the Study - This chapter summarizes key findings, acknowledges limitations, provides policy and investment recommendations, and suggests directions for future research.

CHAPTER TWO

LITERATURE REVIEW

2.1 INTRODUCTION

This chapter covers the theoretical foundations of time series analysis, emphasizing key concepts such as local stationarity, decomposition methods, volatility modeling, and the basis for forecasting financial time series. Econometric models are very close to machine learning, but they are also limited by their applications and forecasting. If you have methodological concerns, there is empirical evidence in international contexts, with the same attention as the study has been conducted in many countries as Ghana. This is complemented by a focused review of research specific to Ghana's treasury bill market, by identifying the uniqueness and challenges of this context. The literature review concludes with a systematic analysis of gaps in existing research, which covers both the methodological limitations and contextual factors specific to Ghana, which has already been talked about enough.

The chapter offers a conceptual framework for forecasting interest rates on Ghanaian public bonds based on these observations. This framework includes a number of theoretical stances and practical approaches. This paradigm serves as the basis for the empirical techniques employed in the following chapters, building upon tried-and-true techniques and addressing recognized research gaps. The evaluation focuses especially on how different approaches are applied in Ghana's economic environment, which is marked by complex interrelationships between fiscal, monetary, and external factors, policy changes, and structural disruptions.

2.2 THEORETICAL FRAMEWORK OF INTEREST RATES AND TREASURY BILLS

2.2.1 Interest Rate Theories

The theoretical underpinnings of interest rate determination and dynamics are crucial for understanding the behavior of Treasury bill rates. The classical loan funds theory posits that interest rates are determined by the supply of and demand for loanable funds in an economy (Kwakye, 2021). This framework continues to provide a fundamental basis for analyzing interest rate movements in economies like Ghana, where the government's borrowing needs significantly influence market rates. Building on this foundation, the liquidity preference theory advanced by Keynes offers additional insights into the determination of short-term interest rates such as Treasury bills. (Boadi & Amegbe, 2017) apply this framework to Ghana's context, demonstrating how preference for liquidity among investors influences the pricing of short-

term government securities, particularly during periods of economic uncertainty. Their analysis showed that liquidity preference intensified during Ghana's 2020 economic downturn, contributing to upward pressure on T-bill rates despite monetary easing.

The expectations theory of the term structure of interest rates provides another important theoretical lens for analyzing Treasury bill rates. According to this theory, long-term interest rates reflect expectations about future short-term rates (Dzigbede & Ofori, 2022). Their empirical investigation of Ghana's yield curve between 2017-2021 found evidence supporting the expectations hypothesis, indicating that current T-bill rates incorporate market expectations about future monetary policy and inflation developments. More recent theoretical development has emphasized the role of institutional factors that affect interest rate determination. (Kumi, 2023) presents an institutional theory of interest rates in developing economies, arguing that the quality of institutions, governance structures, and regulatory frameworks significantly influence interest rate dynamics beyond purely economic factors. This perspective is particularly relevant for Ghana, where institutional reforms in the financial sector have coincided with changes in the behavior of Treasury bill rates.

2.2.2 Treasury Bills as Financial Instruments

Treasury bills represent short-term debt obligations issued by governments, typically with maturities of one year or less. As zero-coupon instruments, treasury bill rates are issued at a discount to their face value, with the difference representing the implied interest earned (Acquah & Khormortey, 2019). In Ghana's financial system, treasury bill rates are issued with standardized maturities of 91 days, 182 days, and 364 days, constituting the most actively traded government securities in the domestic market. (Antwi, Mensah , & Boateng , 2020). (2020) examine the evolving role of Treasury bill rates in Ghana's financial architecture, noting their dual function as monetary policy instruments and investment vehicles. Their analysis highlights how treasury bill serve as a benchmark for pricing other financial assets in Ghana's market, influencing lending rates, corporate bond yields, and even equity valuations through the risk-free rate component.

The pricing mechanism for treasury bill rates in Ghana follows a competitive multiple-price auction system, where market participants submit bids specifying quantities and yields. (Addo, Mensah , & Asante, 2020) provide a detailed analysis of Ghana's treasury bill auction mechanics, demonstrating how the weekly auctions conducted by the Bank of Ghana reveal market sentiment and expectations about economic conditions. Their study finds that auction

outcomes are highly sensitive to market liquidity conditions and fiscal announcements, creating distinct patterns in yield movements.

2.2.3 Ghana's Financial Market and Treasury Bill System

2.2.3.1 Evolution of Ghana's Financial Market

Ghana's financial market has undergone significant transformations over the past decades, evolving from a highly regulated system to a more liberalized and diversified market structure. (Bondzi-Simpson, Aryetey , & Danquah , 2019) documented this evolution, highlighting the financial sector reforms initiated in 1980 and subsequent developments that have shaped the current landscape. Their historical analysis shows how the establishment of the Ghana Stock Exchange in 1990, the introduction of a primary dealer system for government securities in 1996, and the implementation of an electronic book-entry system in 2004 progressively modernized the market infrastructure.

Since, the period of 2017, the witnessed accelerated which reforms in Ghana's financial sector. (Alagidede & Mensah, 2018) analyze the impact of the financial sector cleanup initiated, noting how the resolution of distressed banks and specialized deposit-taking institutions enhanced market stability but temporarily affected liquidity conditions in a treasury bill market. Their findings indicate that the cleanup exercise initially created an upward pressure on treasury bill rates as the restructured financial institutions adjusted their investment portfolios.

Digital transformation has emerged as a defining feature of Ghana's financial market development in recent years. (Agyekum, 2021) examines how technology adoption has influenced market participation and efficiency in Ghana's government securities market. The study highlights the introduction of the Ghana Fixed Income Market (GFIM) electronic trading platform in 2018 and the subsequent implementation of the Central Securities Depository new infrastructure, which enhanced price discovery and liquidity in the treasury bill market.

2.2.3.2 Institutional Framework for Treasury Bills in Ghana

The institutional architecture for managing treasury bill in Ghana involves several key entities. The Ministry of Finance determines the government's borrowing requirements and issues auction guidelines, while the Bank of Ghana conducts the auctions and serves as the registrar and transfer agent for government securities (Quarcoo & Allen, 2020). Their institutional analysis demonstrates how this arrangement creates a complex dynamic between fiscal needs and monetary policy objectives, which is particularly evident during periods of fiscal stress.

Primary Dealers (PDs) play a crucial intermediary role in Ghana's treasury bill market, participating directly in auctions and providing liquidity in the secondary market. (Akosah, Mensah , & Asante, 2022) analyze the evolution of Ghana's primary dealer system, noting how reforms implemented in 2019-2020 strengthened participation requirements and performance monitoring. Their findings suggest that the enhanced primary dealer framework has improved market liquidity but remains challenged by concentration risks, with a small number of dealers dominating trading activity. (Kwakye, Fiscal dominance and interest rate dynamics in Ghana, 2022) examines the complementary role of the Ghana Fixed Income Market (GFIM) as the organized secondary market for Treasury bills and other fixed-income securities. Established in 2015 and significantly enhanced in 2018, GFIM has improved price transparency and market accessibility, though the study identifies persistent challenges in market depth and trading volumes, particularly for short-term instruments like T-bills.

2.2.3.3 MARKET DYNAMICS AND PARTICIPANT BEHAVIOR

Understanding the behavior of different market participants is essential for analyzing treasury bill dynamics in Ghana. (Osei-Assibey & Asenso, 2019) examine the investor base for Ghanaian government securities, noting the dominance of commercial banks, which held approximately 65% of the outstanding treasury bill rate as of 2018. Their analysis reveals how banks' investment strategies and liquidity management practices significantly influence demand patterns and yield movements in the treasury bill market. Foreign investor participation in Ghana's domestic debt market has increased in recent years, introducing new dynamics in the treasury bill market. (Quartey & Gameli, Foreign investor participation and domestic debt market dynamics in Ghana, 2021) analyze the impact of non-resident investors on Ghana's government securities market, finding that increased foreign participation between 2017-2020 contributed to downward pressure on yields but also introduced greater vulnerability to external shocks and capital flow reversals. Their study highlights how the COVID-19 pandemic in 2020 triggered significant portfolio outflows, contributing to yield volatility in the treasury bill market.

Retail investor participation represents another important dimension of Ghana's treasury bill market. (Boateng & Amponsah, 2020) examine the growth of retail investment in treasury bill rates following the introduction of digital platforms that simplified access for individual investors. Their survey-based study reveals how yield considerations dominate investment decisions among retail participants, with perceived safety and availability of electronic channels also serving as significant motivators.

2.3 TIME SERIES FORECASTING: CONCEPTS AND APPLICATIONS

2.3.1 Theoretical Foundations of Time Series Analysis

Treasury bill forecasting is based on time series analysis. This approach essentially involves dissecting a time series into its systematic components (trends, seasonality, and cycles) and random fluctuations to find patterns that can guide forecasts. The toolkit for financial forecasting applications has grown as a result of recent developments in the field. In time series analysis, stationarity is still a crucial idea. The stationarity of financial time series for Ghana, including government interest rates, was investigated by Aientimi et al. (2019). After calculating the first discrepancies, they discovered that these rates were no longer level-stationary. For Ghanaian financial data, their study employed extended unit root analyses, which account for structural fractures and yield more reliable results than simple approaches. To manage the intricate patterns frequently observed in financial time series, decomposition techniques have advanced. The Prophet decomposition approach, developed by Taylor & Letham (2018), has gained popularity in financial forecasting due to its ability to handle non-linear trends and numerous seasonal patterns. Anokye et al. (2021) find weekly, monthly, and quarterly seasonal trends in the 91-day Treasury bill rate series by applying a similar decomposition framework to Ghanaian interest rates.

2.4.2 Traditional Econometric Models

The Autoregressive Integrated Moving Average (ARIMA) method remains central to time series forecasting. Afrane et al. (2022) applied various ARIMA models to Ghana's interest rate data, showing these models are still useful for short-term forecasting. Their analysis showed that ARIMA models performed better than simple forecasts for Ghana's 91-day treasury bill rate over the periods of three months, but became less accurate for longer forecasting periods. Seasonal extensions of ARIMA models (SARIMA) include regular patterns that occur at fixed intervals. Oteng-Abayie et al. (2018) used SARIMA models for Ghanaian financial indicators, and found significant seasonal patterns in treasury bill rates that match the government's borrowing cycle. Their results suggest that accounting for these seasonal patterns improves forecast accuracy, particularly for the 91-day treasury bill rate, which exhibits the strongest seasonality.

For capturing volatility dynamics in financial time series, GARCH models and their variations have proven valuable. Quartey & Ennin (2023) apply various GARCH specifications to model the volatility of Ghana's treasury bill rates, finding evidence of volatility clustering and

asymmetric responses to shocks. Their results indicate that GJR-GARCH models, which account for different impacts of positive and negative shocks, provide a superior fit for Ghana's treasury bill data compared to standard GARCH formulations. Vector Autoregression (VAR) models extend the univariate time series framework to capture dynamic interactions among multiple variables. Lamptey et al. (2020) implement a VAR approach to model the joint dynamics of T-bill rates, inflation, exchange rates, and monetary policy indicators in Ghana. Their findings highlight the importance of accounting for these interactions, as innovations in monetary policy variables and inflation significantly influence subsequent movements in T-bill rates. Exponential Smoothing models, including the Holt-Winters method, are commonly used for short-term financial forecasting. These models give more weight to recent observations, making them effective for detecting trends and seasonality (Holt, 2004). Research by Amponsah et al. (2023) showed that the Holt-Winters method provided good forecasting performance for T-bill rates but had trouble with long-term trend reversals.

2.4.3 Machine Learning Approaches

In recent years, machine learning techniques have been used for financial time series forecasting. Akosah et al. (2023) compared traditional econometric models with various machine learning approaches for forecasting Ghana's interest rates. They found that ensemble methods like Random Forests and Gradient Boosting machines outperform traditional models for longer forecast periods. Their studies suggest that machine learning approaches better capture non-linear relationships in Ghana's interest rate dynamics during periods of economic stress.

Deep learning models, particularly Long Short-Term Memory (LSTM) networks, have shown promise in financial forecasting applications. Osei-Owusu and Awartey (2022) implement LSTM networks for predicting Ghana's T-bill rates, demonstrating their ability to capture long-range dependencies in the data. Their comparative analysis indicates that LSTM models outperform both traditional ARIMA models and simpler machine learning approaches for medium-term forecasting horizons of 3-6 months.

XGBoost, a gradient boosting algorithm, has been widely used in financial forecasting due to its ability to handle missing data and improve prediction accuracy (Chen & Guestrin, 2016). A study by Kwarteng & Opoku (2023) applied XGBoost to forecast Ghana’s interest rates, demonstrating that it performed better than traditional models when combined with macroeconomic indicators. XGBoost is advantageous because it reduces overfitting and is

computationally efficient, but it lacks interpretability compared to statistical models. Kwarteng & Opoku, 2023. Hybrid approaches combining econometric methods and machine learning techniques represent an emerging frontier in forecasting methodology. Adjei-Mantey and Lartey (2024) developed a hybrid forecasting framework that integrates ARIMA components with neural networks, applied to Ghana's T-bill rate series. Their proposed model outperforms standalone approaches in out-of-sample tests, suggesting that combining the strengths of different methodologies can enhance forecasting performance.

2.4.4 PROPHET MODELS FOR TIME SERIES FORECASTING

Prophet was developed by Facebook researchers (now Meta), Taylor & Letham (2018), and implements an additive regression model to decompose time series into trend, seasonal, and holiday components. Unlike traditional models that require extensive parameter tuning, Prophet automates much of the process and can be used even by analysts with limited statistical expertise. Ofori-Boateng et al. (2023) evaluate the performance of Prophet using Ghanaian financial data and find that it is particularly resilient to gaps in the data series, a common problem with financial indicators in developing markets. Their analysis also highlighted Prophet's ability to incorporate external regressors through its additive framework, which allows for the inclusion of macroeconomic indicators such as inflation and exchange rates.

Table 1.Table 1 1 showing the strengths and weaknesses

Model	Strengths	Weaknesses
ARIMA	Well-established, interpretable	Assumes stationarity, limited to nonlinear trends
SARIMA	Captures seasonality	Requires careful parameter tuning
Exponential Smoothing	Good for short-term trends	Struggles with trend reversals
Neural Network	Learns non-linear relationships	Performs poorly on small datasets
Prophet	Additive model with trend, seasonality, and holiday components	Handles missing data, automatic changepoint detection, and flexible seasonality

2.5 PREVIOUS STUDIES

2.5.1 INTERNATIONAL EVIDENCE

A substantial body of literature exists on forecasting interest rates and government securities yields in various countries, which provides valuable methodology insights that apply to the Ghanaian context. Fakoya et al. (2022) conducted a comprehensive review of interest rate forecasting studies across emerging markets, which analyzed the evidence on model performance and the methods approach. Their meta-analysis showed that country-specific factors affect model selection and forecast accuracy, which highlights the importance of theoretical considerations in forecasting applications.

The are comparative studies that examine the performance of different market environments that offer particularly relevant insights. Nguyen et al. (2020) analyzed treasury bill rate forecasting in Vietnam, Malaysia, and Thailand, concluding that model performance varies systematically with market development stage and macroeconomic stability. Their results suggest that simpler models often perform better in less developed markets with higher volatility, a finding potentially relevant for Ghana's context. Recent innovations have expanded the forecasting toolkit available for interest rate applications. Kim et al. (2023) introduce a quantum-inspired optimization approach for calibrating treasury bill forecasting models, by demonstrating better improvements over conventional optimization methods. Their framework, tested on data from multiple emerging markets, including Ghana, suggests that advances in computational methods can enhance forecasting performance even with an established model structure.

The COVID-19 pandemic created unprecedented challenges for interest rate forecasting globally. Osoro et al. (2022) examine forecasting performance during the pandemic across several African economies, which found that models incorporating external factors and policy response variables demonstrated greater resilience than univariate approaches. Their analysis of Ghana's case highlights how extraordinary monetary policy interventions during the pandemic created structural breaks in the treasury bill rate series, complicating forecasting efforts.

2.5.2 STUDIES SPECIFIC TO GHANA

Research specifically focused on forecasting Ghana's Treasury bill rates has expanded in recent years, providing insights into the unique characteristics of the Ghanaian market. Agyekum et al. (2020) apply various ARIMA specifications to Ghana's 91-day treasury bill rate, finding that an ARIMA (2,1,2) model provides the best fit for their sample period of 2014-2019. Their

analysis identifies significant autocorrelation patterns in the differenced series, indicating determined memory in Ghana's treasury bill rate dynamics.

Studies assessing multiple modeling approaches provide valuable evidence on relative performance. Donkor & Danso (2019) evaluate ARIMA, exponential smoothing, and neural network models for forecasting Ghana's 91-day and 182-day treasury bill rates. it was found out that neural networks outperform other approaches for longer forecast horizons but underperform ARIMA models for short-term predictions. Their results suggest that the appropriate modeling approach depends critically on the intended forecasting horizon.

The relationship between macroeconomic variables and treasury bill rates in Ghana has been examined by several researchers. Baidoo and Akoto (2021) investigate the predictive power of macroeconomic indicators for Ghana's treasury bill rates movements, finding that inflation, exchange rates, and monetary policy rates exhibit significant predictive content. Their findings suggest that incorporating these variables through transfer function models or multivariate approaches can enhance forecasting performance compared to univariate specifications.

Structural breaks represent a significant challenge for time series modeling in Ghana's context. Aboagye et al. (2021) identify multiple structural breaks in Ghana's treasury bill rate series, corresponding to major policy changes and external shocks, and evaluate forecasting approaches that explicitly account for these discontinuities. Their results indicate that models allowing for regime switching or incorporating structural break indicators outperform standard specifications that assume parameter stability.

2.6 GAP ANALYSIS IN EXISTING LITERATURE

2.6.1 METHODOLOGICAL GAPS

Despite the growing literature on interest rate forecasting in Ghana, several methodological gaps remain. First, as noted by Ampah and Karikari (2021), most existing studies rely on linear modeling approaches despite evidence of non-linearity in Ghana's interest rate approach. Their analysis explains the threshold effects and asymmetric adjustments in treasury bill rates, suggesting a potential value of non-linear modeling approaches that remain underexplored in the Ghanaian context. Many applications of machine learning and deep learning methods to Ghana's interest rate forecasting remain limited compared to international applications. While recent studies like Osei-Owusu & Awartey (2022) have begun to address this gap, comprehensive comparisons of traditional econometric and advanced machine learning approaches across different market conditions and time horizons are still scarce.

Most existing studies focus on point forecasts without adequate consideration of forecast uncertainty. Alagidede & Kuranchie (2021) highlight this limitation, noting how uncertainty

quantification becomes particularly important in volatile market environments like Ghana's. Their preliminary work on probabilistic forecasting of Ghana's interest rates demonstrates the potential value of approaches that generate prediction intervals or density forecasts rather than single-point estimates. The treatment of structural breaks and regime changes in forecasting applications remains underdeveloped. While studies like Aboagye et al. (2021) have identified these features in Ghana's treasury bill rate series, systematic approaches for incorporating regime changes in forecasting models remain limited. This gap is particularly significant given Ghana's history of policy shifts and external shocks that create discontinuities in financial time series.

2.6.2 CONTEXTUAL AND EMPIRICAL GAPS

Beyond methodological considerations, several contextual and empirical gaps exist in the literature on Ghana's treasury bill forecasting. First, the role of market microstructure and institutional factors in treasury bill rate dynamics remains unexplored. Peprah & Boateng (2023) highlight this gap, noting how auction mechanisms, primary dealer behavior, and regulatory changes influence treasury bill yields in ways not captured by macroeconomic variables alone. Their preliminary analysis suggests that incorporating market structure variables could enhance forecasting models, but comprehensive empirical work in this direction remains limited.

The interaction between domestic and external factors in driving Ghana's treasury bill rates requires further investigation. Gyeke-Dako et al. (2022) examine how global financial conditions and capital flows influence Ghana's domestic interest rates, finding significant spillover effects that vary across different market regimes. However, the incorporation of these external dynamics into forecasting models remains underdeveloped, with most existing approaches focusing primarily on domestic variables.

The forecasting implications of Ghana's fiscal dominance, where fiscal considerations significantly influence monetary outcomes, remain inadequately addressed. (Kwakye, Fiscal dominance and interest rate dynamics in Ghana, 2022) highlights how Ghana's persistent fiscal deficits and high government borrowing needs create distinct interest rate dynamics that may require specialized modeling approaches. The development of forecasting models that explicitly account for fiscal-monetary interactions represents an important direction for future research. the temporal stability of forecasting relationships in Ghana's context requires more systematic investigation. While several studies have documented changing patterns in Ghana's financial markets, the implications for model selection and forecasting strategy across different

economic environments remain unclear. This gap is particularly relevant given Ghana's recent experiences with economic volatility and policy transitions.

2.6.3 CONCEPTUAL FRAMEWORK

Building on the reviewed literature and identified gaps, this study adopts a conceptual framework that integrates multiple theoretical perspectives and methodological approaches for forecasting Ghanaian fiscal accounts. This paradigm takes into account the difficulty of creating fiscal policy by examining the interplay of monetary policy, fiscal dynamics, market structure, and outside factors. However, it also tackles the methodological issues caused by non-linearities, structural distortions, and regime changes. This method is based on the knowledge that Ghana's fiscal reports contain time series aspects that require careful modeling. These consist of non-stationarity, seasonality, cumulative volatility, and possible structural flaws. The framework uses univariate time series techniques to capture these features, but it also acknowledges the potential advantages of multivariate models that account for interactions with macroeconomic variables and external influences. The framework explicitly acknowledges the potential complementarity between traditional econometric approaches and machine learning methods, positioning these not as competing alternatives but as tools with different strengths that may be combined for enhanced forecasting performance. This perspective aligns with recent work by Amoah et al. (2023), who demonstrate the value of model combination and ensemble approaches for financial forecasting in Ghana's context. The conceptual framework also incorporates uncertainty quantification as an essential component of the forecasting process, recognizing that point forecasts alone provide incomplete information for decision-making, particularly in volatile environments. This element addresses the gap identified by Alagidede & Kuranchie (2021) regarding the limited attention to forecast uncertainty in existing studies.

Finally, the framework acknowledges the dynamic nature of forecasting relationships, allowing for time-varying parameters and adaptive modeling approaches that can adjust to changing market conditions. This component addresses the temporal stability concerns highlighted in the gap analysis and aligns with recent work by Agyekum and Doku (2022) on adaptive forecasting techniques for Ghanaian financial markets. This integrated conceptual framework guides the methodological choices and empirical strategy of the current study, addressing key gaps in the existing literature while building on established approaches. The framework provides a structure for comparing alternative forecasting methods, evaluating their performance across

different market conditions, and developing recommendations for practical applications in policy and investment contexts.

2.7 SUMMARY OF THE CHAPTER

This chapter provides a comprehensive literature review on time series forecasting with a particular focus on Treasury bill interest rates in Ghana. The analysis began with a review of the theoretical foundations of time series analysis and highlighted key concepts such as stationarity, decomposition methods, and volatility modeling. Traditional econometric approaches and emerging machine learning techniques were evaluated, and their respective strengths and weaknesses in financial forecasting were highlighted. The gap analysis identified several areas requiring further research. Methodologically, there is still a need for more systematic research into nonlinear modeling methods, a comprehensive comparison of traditional and machine learning methods, improved uncertainty measurement, and better management of structural attrition. From a contextual perspective, the deficiencies include the limited consideration of microstructural market factors, the incomplete modeling of endogenous and exogenous interactions, the inadequate treatment of financial dominance effects, and the inadequate consideration of the time stability of forecast relationships.

A conceptual framework that incorporates different theoretical viewpoints and methodological techniques for forecasting interest rates on Ghanaian treasury bills is created in light of these findings. By acknowledging the possible overlap between machine learning and conventional econometric techniques, this framework tackles the intricate process of determining treasury interest rates through the interplay of monetary policy, fiscal dynamics, market structure, and external influences. The literature review thus provides a solid foundation for the empirical analysis to follow, highlighting both established methodologies and emerging approaches that can contribute to improved forecasting of Ghana's treasury bill rates. By addressing the identified gaps and building on existing research, this study aims to enhance understanding of interest rate dynamics in Ghana's context and provide practical insights for policy and investment applications.

CHAPTER THREE

METHODOLOGY

3.1 RESEARCH DESIGN

This study employs a quantitative research design with an emphasis on time series forecasting to predict treasury bill rates in Ghana. In the study of Ampomah & Mensah (2021), quantitative approaches are particularly suitable for financial market research where objective measurement and statistical analysis are required. The research follows a deductive approach, testing and establishing time series forecasting models on Ghana treasury bill data to determine their predictive accuracy. This study adopts a comparative methodological framework, evaluating both traditional econometric time series models and contemporary machine learning techniques. This dual approach enables a comprehensive assessment of forecasting performance across different modeling approaches. This research design incorporates both in-sampling model fitting and out-of-sample forecast evaluation to ensure robust performance assessment. The time horizon for the study spans from June 2017 to April 2025, providing a sufficient timeframe to capture various economic policy and market conditions in Ghana, which allows more reliable parameter estimation and model validation across different economic environments.

3.2 DATA SOURCE AND PREPROCESSING

3.2.1 DATA SOURCE

The primary data for this study consists of historical treasury bill rates for 91-day instruments in Ghana. The rate will be collected from the Bank of Ghana website. The Bank of Ghana represents the most reliable and authoritative source for Ghana treasury bill rate data, which ensures data integrity and consistency.

3.2.2 DATA PREPROCESSING

Before the model estimation, the data undergoes several preprocessing steps:

- Missing Value Treatments: Missing observations in the dataset for the time series are identified and addressed through appropriate imputation methods. Linear interpolation is used based on neighboring data for isolated missing values. The study uses more complex imputation approaches for longer periods, like seasonal decomposition methods that take the series' overall trend into account.

- Data Transformation: Appropriate transformations, such as first differencing (for trend elimination) and logarithmic transformation (for variance stabilization), are taken into consideration in order to handle potential non-stationarity and non-normality in the Treasury bill rate series. Both statistical testing and visual examination of the transformed series are used to determine the ultimate transformation strategy.
- Data Splitting: The dataset is split into two parts: 30% for testing and 70% for training. Model estimation is done on the training set, model selection and hyperparameter tuning are done on the validation set, and final model evaluation is done on the testing set. This section makes certain that the forecasting models are assessed using data that wasn't used during their creation.

3.3 STATISTICAL TESTS AND PRELIMINARY ANALYSIS

Before model estimation, several diagnostic tests will be conducted to understand the properties of the time series and ensure appropriate model specification:

3.3.1 STATIONARY TEST

The Augmented Dickey-Fuller test will be employed to assess stationarity. In statistics, the Augmented Dickey-Fuller (ADF) test the null hypothesis that a unit root is present in a time series sample. The alternative hypothesis depends on which version of the test is used, but it is usually stationarity or trend-stationarity. It is an augmented version of the Dickey-Fuller test for a larger and more complicated set of time series models. The augmented Dickey-Fuller (ADF) statistic used in the test is a negative number. The more negative it is, the stronger the rejection of the hypothesis that there is a unit root at some level of confidence.

The procedure for the ADF test is the same as the Dickey-Fuller test, but it is applied to the model:

$$\Delta \boldsymbol{\gamma t} = \boldsymbol{\alpha} + \boldsymbol{\beta t} + \boldsymbol{\gamma y_{t-1}} + \boldsymbol{\delta_1 \Delta y_{t-1}} + \cdots + \boldsymbol{\delta_{p-1} \Delta y_{t-p+1}} + \boldsymbol{\varepsilon_t},$$

Where α is a constant, β the coefficient on a time trend and ρ The lag order of the autoregressive process. Imposing the constraints $\alpha = 0$ and $\beta = 0$ corresponds to modelling a random walk and using the constraints $\beta = 0$ Corresponding to modelling a random walk with a drift. Consequently, there are three main versions of the test, analogous to those of the Dickey-Fuller test.

3.3.2 VISUAL PLOTS

- Time Series Plots: To identify trends, seasonality, and potential structural breaks in the data.
- Seasonal Plots: To assess the presence and pattern of seasonality in Treasury bill rates.
- Autocorrelation Function (ACF): measures the relationship between a time series and a lagged version of itself over successive time intervals. Simply put, it tells you how similar the data points are to each other at different time lags. With autocorrelation, there are two different plot strategies to measure the functions. The autocorrelation function and partial autocorrelation function. The autocorrelation function plots the correlation of the time series with itself at different lags. This helps in identifying patterns such as seasonality, trends, and the persistence of values over time.
- Partial Autocorrelation Function (PACF): measures the correlation between observations at two time points, which accounts for the values of the observations at all shorter lags. This helps isolate the direct relationship between observations at different lags, removing the influence of intermediary observations.

3.4 TIME SERIES FORECASTING MODELS

This study aims to employ a range of time series models, including traditional econometric and machine learning techniques. Each model category offers distinct advantages and is suited to capturing different aspects of time series dynamics.

3.4.1 ARMA Model

The Autoregressive Moving Average model is a mixture autoregressive moving average model of order 1, ARMA (1,1), if it satisfies the following equation:

$$\boldsymbol{Xt} = \mu + \boldsymbol{\varphi Xt - 1} + \boldsymbol{\epsilon_t} + \boldsymbol{\theta \epsilon_{t-1}} \, \forall \boldsymbol{t}$$

$$\boldsymbol{\Phi(L)Xt} = \mu + \boldsymbol{\Theta(L)\epsilon_t}$$

Where $\boldsymbol{\theta \neq 0, \theta \neq 0,}$ $\boldsymbol{\mu}$ is a constant term, $\boldsymbol{(\epsilon_t)_{t \in \mathbb{Z}}}$, is a weak white noise process with expectation zero and variance $\boldsymbol{\sigma_\epsilon^2}$ $\left(\boldsymbol{\epsilon_t \sim WN(0, \sigma_\epsilon^2)}\right)$,

$$\boldsymbol{\Phi(L) = 1 - \varphi L \; and \; \Theta(L) = 1 + \theta L}.$$

3.4.2 ARIMA Model

The ARIMA (Autoregressive Integrated Moving Average) is a common time series forecasting model. It's a more sophisticated extension of the ARMA model, with the addition of an

integration component. ARIMA is a non-seasonal model, which is usually denoted as ARIMA (p, d, q) where the parameters p, d, and q are non-negative integers: p is the order of the autoregressive model, d is the degree of differencing, and q is the order of the moving average model. Seasonal ARIMA models are also usually denoted ARIMA (p, d, q) (P, D, Q)m, where the upper cases are the autoregressive, differencing, and moving average terms for the seasonal part of the ARIMA.

- AR: Autoregressive captures the relationship between an observation and a specified number of lagged observations.
- I: Integrated represents the differencing of observations to make the time series stationary.
- MA: Moving Average in this aspect of the model, it's models the error term as a linear combination of error terms occurring contemporaneously and at various times in the past.

$$\left(1 - \sum_{i=1}^{p} \varphi i L^i\right)(1 - L)^d X_t = \delta + \left(1 + \sum_{i=1}^{q} \theta_i L^i\right)\varepsilon_t$$

The model-building process follows the Box-Jenkins methodology:

- Identification: Determining appropriate values for p, d, and q through ACF/PACF analysis and information criteria
- Estimation: Estimating the model parameters using maximum likelihood estimation
- Diagnostic checking: Verifying model adequacy through residual analysis
- Forecasting: Generating predictions based on the fitted model.

3.4.3 SARIMA Model

The SARIMA (Seasonal ARIMA) is an extension of the ARIMA model, incorporating seasonality into its structure.

- Seasonal AR captures the autoregressive nature of the data at the seasonal lag.
- Seasonal I differencing at the seasonal intervals, not just at the first lag.
- Seasonal MA, the moving average components for the seasonal part of the time series.

$$(1 - \phi_1 B)(1 - \Phi_1 B^4)(1 - B)(1 - B^4)y_t = (1 - \theta_1 B)(1 + \Theta_1 B^4)\epsilon_t$$

So, SARIMA models are denoted as SARIMA (p, d, q) (P, D, Q) s where:

- p is the number of autoregressive terms
- d is the number of nonseasonal differences needed for stationarity
- q is the number of lagged forecast errors in the prediction equation
- P, D, and Q are the seasonal equivalents of p, d, and q
- S is the length of the seasonal cycle

3.4.4 Exponential Smoothing

Exponential smoothing, sometimes referred to as exponential moving average (EMA), is a general technique for smoothing time series data that makes use of the exponential window function. While the basic moving average weights the previous observations equally, exponential functions apply exponentially diminishing weights over time. The procedure is straightforward, easy to understand, and can be used to make a variety of judgments based on the user's assumptions, including seasonality. In time-series data analysis, exponential smoothing is commonly used.

The raw data sequence is often represented by $\{x_t\}$ beginning at a time $\boldsymbol{t = 0}$, and the output of the exponential smoothing algorithm is commonly written as $\{\boldsymbol{s_t}\}$, which may be regarded as a best estimate of what the next value of x will be. When the sequence of the observations begins at time *t = 0,* exponential smoothing, which is simple, is given by these formulas:

$$\boldsymbol{s_0 = x_0}$$

$$\boldsymbol{s_t = \alpha x + (1 - \alpha) s_{t-1}, t > 0}$$

Where α is the smoothing factor, and $0 < \alpha < 1$. If $\boldsymbol{s_{t-1}}$ is substituted into $\boldsymbol{s_t}$ continuously so that the formula of $\boldsymbol{s_t}$ is fully expressed in terms of $\boldsymbol{x_t}$ then exponentially decaying weighting factors on each raw data $\boldsymbol{x_t}$ is revealed, showing how exponential smoothing is named.

3.5 Machine Learning Model

Neural networks form the base of deep learning, which is a subfield of machine learning. The structures of the human brain inspire the algorithms. A neural network takes input data, trains itself to recognize patterns found in the data, and then predicts the output for a new set of similar data. Therefore, a neural network can be classified as the functional unit of deep learning, which mimics the behavior of the human brain to solve complex problems, or a neural network model can be defined by a function that takes an input (observation) and produces an output (decision) $f \rightarrow Y$ or distribution over both X and Y. A neural network is essentially a thick network of interconnected layers. These layers are further divided into basic units called

perceptron's, which are fundamentally derived from neurons. The processing unit, input terminals, and output terminals make up a perceptron. A perceptron's input terminals are linked to the output terminals of its predecessors. The process of passing data through the neural network is known as forward propagation, and it occurs in a perceptron, which is a basic unit of a neural network. The process can be broken down into three simple steps:

Step 1: For each input, multiply the input value x_i with weights w_i and sum all the multiplied values. The weight represents the strength of the connection between the neurons and determines how much influence the given input will have on the neuron's output. Input x_1 will have more impact on the output than the weight w_2 if the weight w_1 has a higher value than the weight w_2.

$$\sum = (\boldsymbol{Xi} \times \boldsymbol{Wi}) + (\boldsymbol{Xj} \times \boldsymbol{Wj}) + \cdots + (\boldsymbol{Xn} \times \boldsymbol{Wn})$$

The row vectors of the inputs and weights are $\boldsymbol{X} = [\boldsymbol{Xi}, \boldsymbol{Xj}, \ldots, \boldsymbol{Xn}]$ **and** $\boldsymbol{W} = [\boldsymbol{Wi}, \boldsymbol{Wj}, \ldots, \boldsymbol{Wn}]$ and their multiply given by

$$\boldsymbol{X} \cdot \boldsymbol{W} = (\boldsymbol{Xi} \times \boldsymbol{Wi}) + (\boldsymbol{Xj} \times \boldsymbol{Wj}) + \cdots + (\boldsymbol{Xn} \times \boldsymbol{Wn})$$

Therefore, the summation is equal to the multiplication of X and W.

$$\sum = \boldsymbol{X} \cdot \boldsymbol{W}$$

- Step 2: A bias is then added to the summation which is multiplied and equal to z. The constant is added to the product of inputs and weights. It is used to offset the result, and it helps the models to shift the activation function towards the positive or negative side.

$$\boldsymbol{Z} = \boldsymbol{X} \cdot \boldsymbol{W} + \boldsymbol{B}$$

- Step 3: The value $\mathcal{Z}$ is then also passed to an activation function. This function introduces non-linearity into the system and helps the neural network learn complex patterns and relationships. The output of the activation function becomes the output of the perceptron. However, perceptron has a binary step function as its activationType equation here. function. Therefore, the activation function that will be used is the sigmoid function or the logistic function.

$$\hat{Y} = \boldsymbol{\sigma}(\mathbf{z})$$

Where σ is denoted as the sigmoid activation function and the $\hat{Y}$ is the forward propagation of the predicted output.

3.5.1 Learning Algorithm

The learning algorithm has two different parts: Backpropagation and Optimization.

Backpropagation of errors refers to the formula used to determine how the loss function's gradient changes about the weights. The following two phases illustrate how a perceptron performs backpropagation.

Step 1: To know the estimation loss function is used. Also, mean squared error is chosen as the loss function for the regression problems. The regression problem and the loss function mean squared error will square the difference between the original ($\mathcal{Y}i$) and predicted value(ŷi).

$$\mathcal{MSE}i = (\mathcal{Y}i - \hat{y}i)^2$$

The loss function is calculated for the entire training dataset and their average which is the cost function.

$$\mathbf{C} = \mathbf{MSE} = \mathbf{1/n} \sum_{i=1}^{n} (\mathcal{Y}i - \hat{y}i)^2$$

- Step 2: To find the best weights and bias for the perceptron, for this to occur, there must be a change in the cost function concerning both weights and bias. This is done with the help of the gradients (rate of change) of how one quantity changes concerning another quantity. Calculate the gradient of cost function C concerning the weight w_i using partial derivation.

$$\frac{dC}{dw_i} = \frac{dC}{d\hat{Y}} \times \frac{d\hat{Y}}{dz} \times \frac{dz}{dw_i}$$

The three gradients are as follows,

Let's start with the gradient of the cost function (C) concerning the predicted value (Ŷ).

$$\frac{dC}{d\hat{Y}} = \frac{d}{d\hat{Y}} \frac{1}{n} \sum_{i=1}^{n} (\mathcal{Y}i - \hat{y}i)^2$$

$$= 2 \times \frac{1}{n} \sum_{i=1}^{n} (\mathcal{Y}i - \hat{y}i)$$

Let $\mathbf{y} = [y\mathcal{i}, y\mathcal{j}, \ldots y_n]$ and $\hat{\mathbf{y}} = [\hat{y}\mathcal{i}, \hat{y}\mathcal{j}, \ldots \hat{y}_n]$ be the row vectors of actual and predicted values.

$$\frac{2}{n} \times \mathbf{sum}\,(\mathbf{y} - \hat{y})$$

Also, the gradient of the predicted value (ŷ) concerning (z).

$$\frac{d\hat{y}}{dZ} = \frac{d}{dZ}\sigma(z)$$

$$= \frac{d}{dZ}\left(\frac{1}{1 + e^{-z}}\right)$$

$$= \frac{e^{-z}}{(1+e^{-z})^2}$$

$$= \frac{1}{(1+e^{-z})} \times \left(1 - \frac{1}{(1+e^{-z})}\right)$$

$$= \sigma(z) \times \left(1 - \sigma(z)\right)$$

Also, the gradient of z concerning the weight ($\mathcal{W}i$).

$$\frac{dZ}{d\mathcal{W}i} = \frac{d}{d\mathcal{W}i}(z)$$

$$= \frac{d}{d\mathcal{W}i} \cdot \sum_{i=1}^{n}(\mathcal{X} \cdot \mathcal{W} + b)$$

$$= \mathcal{X}i$$

Therefore,

$$\frac{dC}{d\mathcal{W}i} = \frac{2}{n} \times \mathbf{sum}\,(y - \hat{y}) \times \sigma(z) \times (1 - \sigma(z)) \times \mathcal{X}i$$

But the bias is theoretically having an input of 1. Hence,

$$\frac{dC}{d\mathcal{b}} = \frac{2}{n} \times \mathbf{sum}\,(y - \hat{y}) \times \sigma(z) \times (1 - \sigma(z))$$

Optimization: The goal of optimization is to choose the optimal element from a set of available choices; in this example, that would be the best perceptron weights and bias.

- Step 1: Choose the gradient descent value that has the optimization algorithm, which will change both the weights and bias, proportional to the negative of the gradient of the cost function to the corresponding weight and bias.
- Step 2: The Learning rate would be used to control the number of times the weights and bias would be changed.
- Step 3: The backpropagation and gradient descent are repeated until convergence.

$$\mathcal{W}i = \mathcal{W}i - \left(\alpha - \frac{dC}{d\mathcal{W}i}\right)$$

$$\mathcal{B} = \mathcal{B} - \left(\alpha - \frac{dC}{d\mathcal{B}}\right)$$

3.5.2 Prophet Model

The Facebook prophet model, specifically for time series, has a strong seasonal pattern and multiple seasonality. It is also implemented by:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

Where:

$g(t)$ is the trend component.

$s(t)$ It is a seasonal component.

$h(t)$ captures holiday effects

Prophet's flexibility in modeling different components makes it potentially valuable for capturing the complex patterns in Treasury bill rates influenced by fiscal cycles and policy changes.

3.6 MODEL SELECTION AND EVALUATION

3.6.1 Model Selection Criteria

- Akaike Information Criteria (AIC)
- Bayesian Information Criteria (BIC)

These criteria balance model fit against complexity to prevent overfitting.

3.6.2 Evaluation Metrics

- Mean Absolute Error (MAE) = $\frac{1}{n} \sum_{t=1}^{n} |yt - \hat{y}t|$
- Root Mean Square Error (RMSE) = $\sqrt{\frac{1}{n}\sum_{t=1}^{n}(yt - \hat{y}t)^2}$
- Mean Absolute Percentage Error (MAPE) = $\frac{100\%}{n}\sum_{t=1}^{n}\left|\frac{yt - \hat{y}t}{yt}\right|$

3.7 SUMMARY

This chapter has outlined the methodological approach for forecasting Ghana's 91-day treasury bill rates. The research is designed to incorporate both traditional econometric models and machine learning approaches, implemented within a rigorous framework of data preprocessing, model selection, and forecast evaluation. This comprehensive methodology aims to identify the most effective forecasting approaches while accounting for the unique characteristics of Ghana's financial market environment.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 INTRODUCTION

This chapter presents the results obtained from analyzing the 91-day treasury bill rate in Ghana. The analysis follows the methodology outline, which employs various time series forecasting models to predict future trends in Ghana's treasury bill rates. This chapter begins with an analysis of the descriptive statistics, data examination, stationarity tests, model identification, diagnostic checking, and forecasting performance evaluation. The results are presented systematically with appropriate interpretations to provide insights into the behavior of Ghana's 91-day treasury bill rate.

4.2 Descriptive Statistics

The analysis covers the 91-day treasury bill rate from June 2017 to May 2025, with over 381 daily observations.

Table 4.1 Table 4 1 showing the data summary

Data Summary	
count	381
mean	18.980523
std	6.971173
min	12.390300
25%	13.413300
50%	14.707800
75%	25.456800
max	36.182500

Interpretation

The mean interest rate is about 18.98%, with a standard deviation of 6.9, and a spread between 13.41% at 25% of the data, and 25.46% at 75% of the data. The data shows that the interest rates are quite variable, and the distribution of the interest rates is skewed to the right.

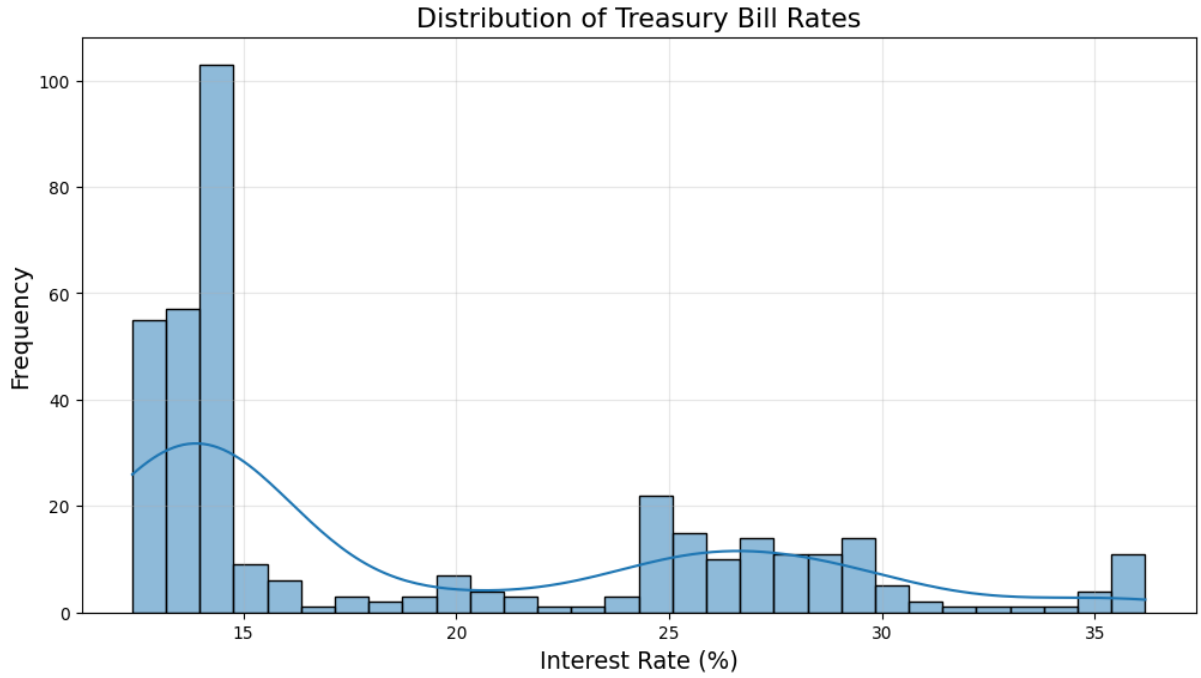


Figure 4.1 showing the distribution of Treasury Bill Rates

Interpretation

The above histogram shows how skewed it is to the right, indicating how evenly spread out they are, which concentrates at the lower end with a long tail to the right.

4.2 Time Series

4.2.1 Time Series Plot

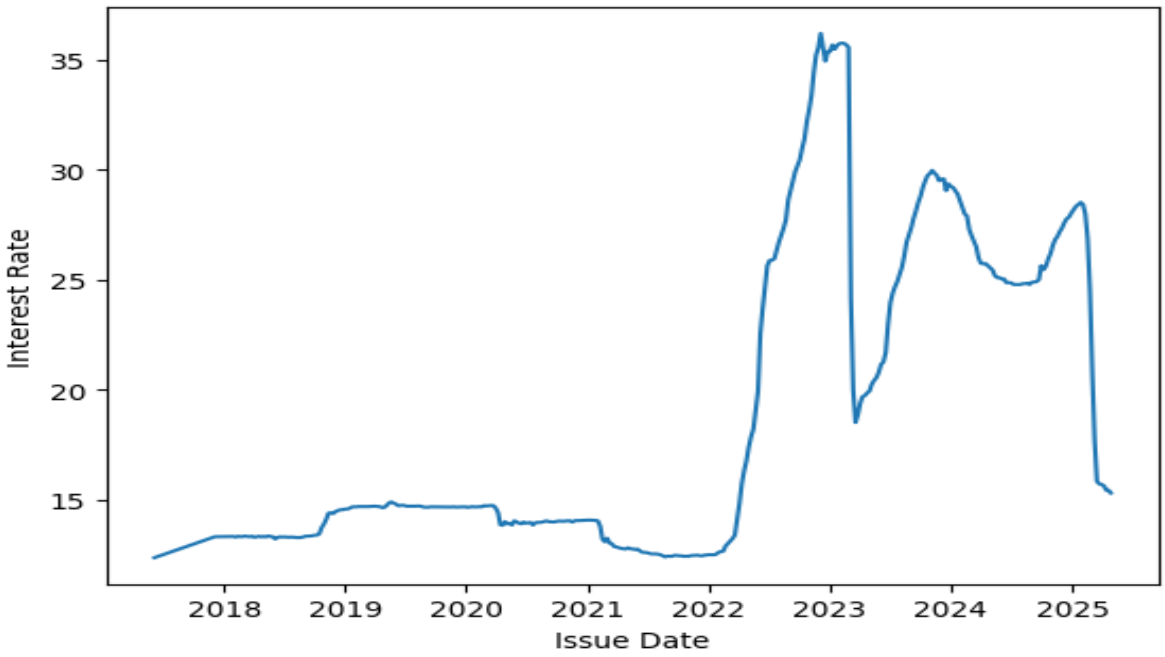


Figure 4 2 showing the time series plot

Figure 4.2

The interest rate was relatively stable, most between 13% and 15% in the year 2017 to mid-2021, which suggested economic stability or consistent monetary policy.

4.2.2 Decomposition

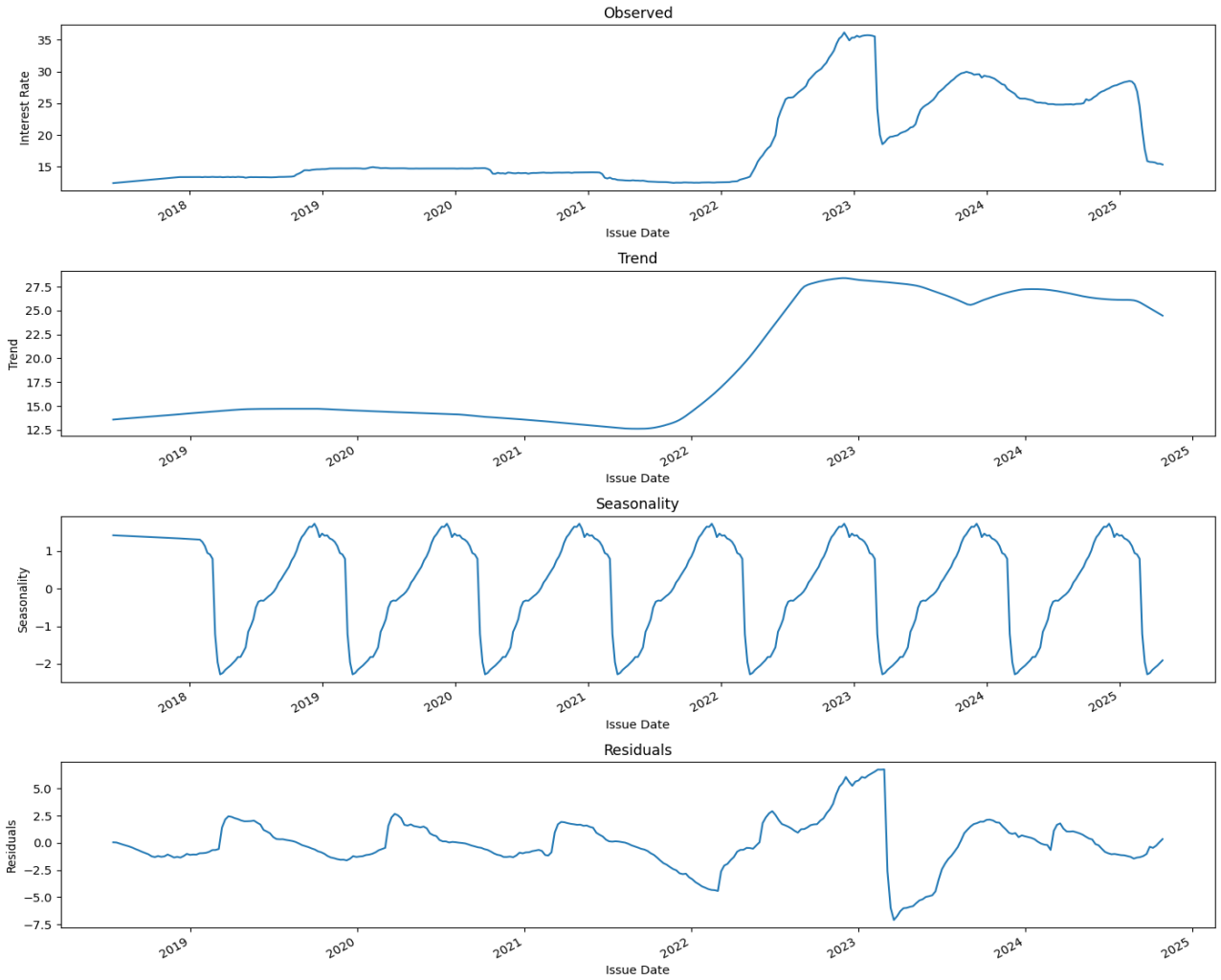


Figure 4.3Figure 4 3 showing the decomposition

Interpretation

The trend line shows a long-term direction of the interest rates and removes short-term fluctuations. The trend increased in 2022, slightly declined in 2023, but remains elevated compared to pre 2022 level.

4.2.2 Autocorrelation Analysis

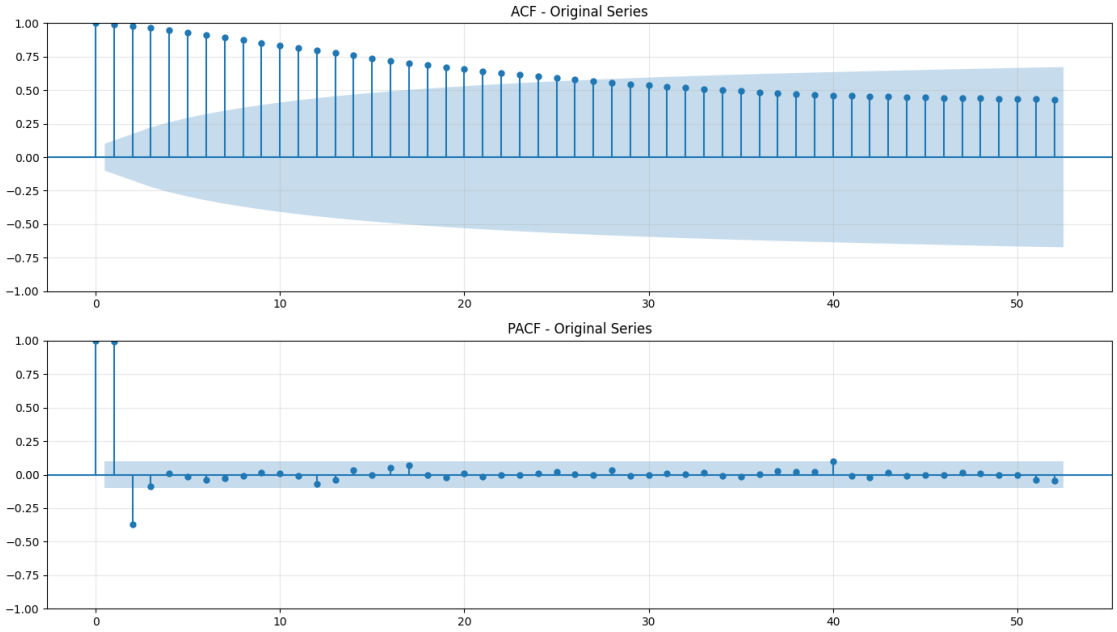


Figure 4.4 Figure 4 4 showing the autocorrelation analysis

Interpretation

The above ACF graph shows strong, slowly decreasing correlations which it starting at 1.0. The patterns show that the ACF has a trend, and since the ACF tails off slowly like this, this means the series is not stationary. While the PACF shows a sharp drop after lag 1, and only occurs at lag 1. This means that the direct relationship in the series is explained by the first lag.

4.2.3 Stationary Test

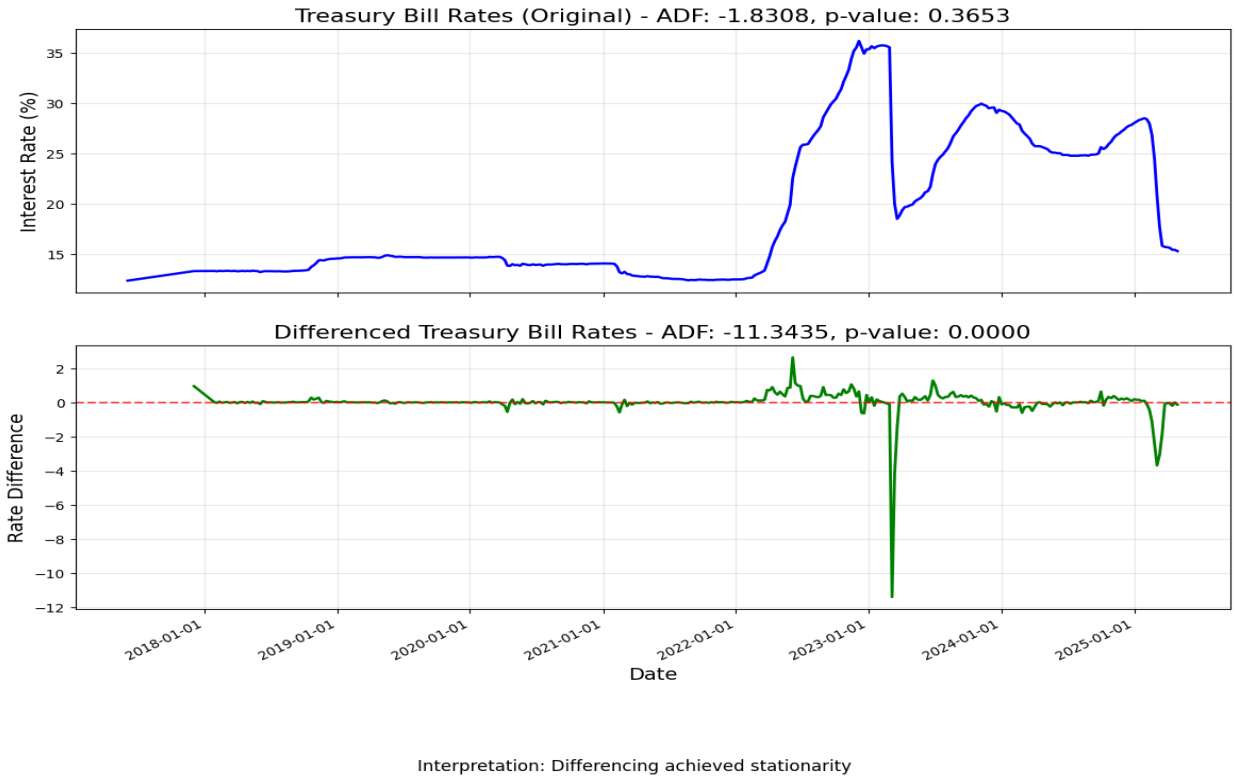


Figure 4.5 Figure 4 5 showing the stationary test

Table 4.2 Table 4 2 showing results of the dickey fuller tests

Dickey-Fuller Tests

Statistic	First Test	Second Test
Test Statistic	-1.830814	-1.134350e+01 (-11.34)
p-value	0.365274	1.040604e-20 (≈0)
Lags Used	1	0
Number of Observations Used	379	379.
Critical Value (1%)	-3.447722	-3.447722
Critical Value (5%)	-2.869196	-2.869196
Critical Value (10%)	-2.570849	-2.570849

Interpretation

The ADF test for the first test is less negative, more than all the critical values, and the p-values exceed the threshold of 0.05 at a 95% confidence interval. Fail to reject the null hypothesis, indicating that the first test is not stationary. While the second test is highly negative, with a p-value less than 0.05, the null hypothesis can be rejected since the test is now stationary.

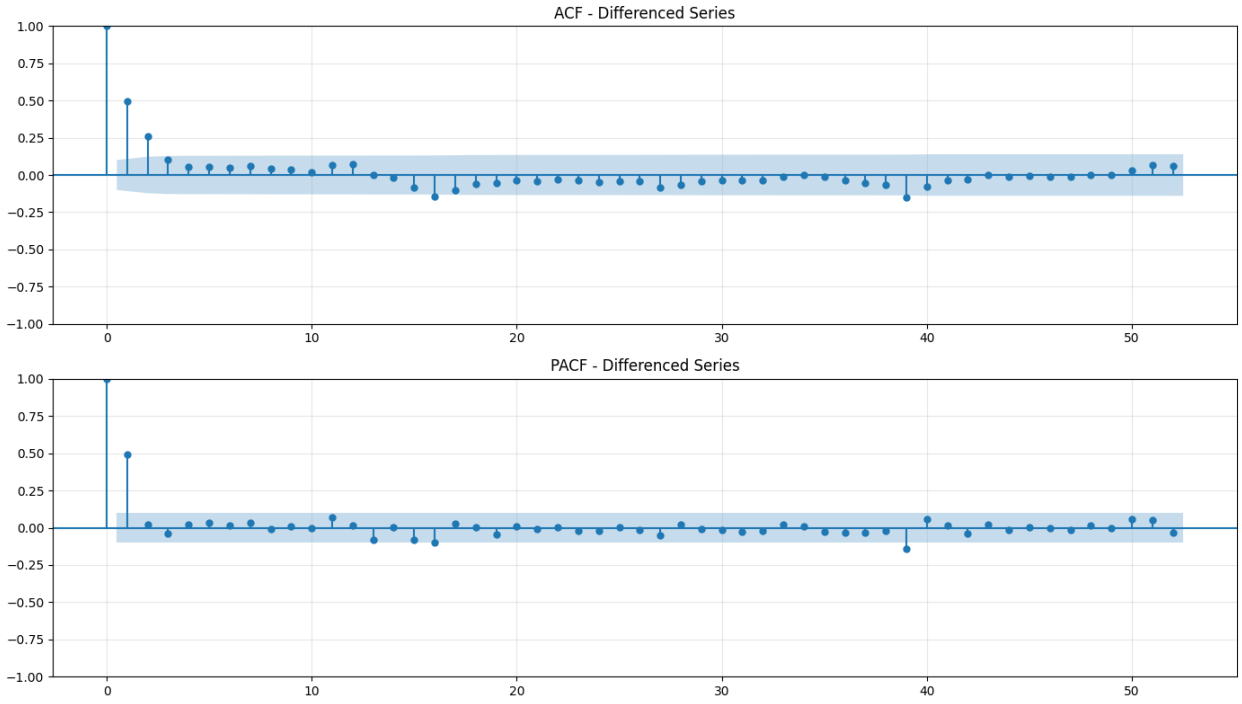


Figure 4.6 Figure 4 6 showing the ACF and PACF graphs

Interpretation

From the above previous ACF graph was a strong correlation at lag 1, while the difference series showed a strong correlation at lag 0, which is around 0.5, with most correlation falling within the significance bands. The rapid decay in the autocorrelation indicates that the differencing has successfully removed the constant trend in the original series. Also, the PACF was significantly positively correlated at lag 1, and most of the other partial autocorrelations fall within the significance bands.

4.3 ARIMA Model

Table 4.3 Table 4 3 showing the ARIMA results

4.3.1 ARIMA (1, 1, 0)

Term	Values
Log Likelihood	-378.97
AIC (761.95)	761.95

BIC	769.83
HQIC	765.08

The ARIMA (1,1,0) model fits the data perfectly well, and it captures the trend after differencing. The AR (1) term is statistically significant, indicating that past values help explain current changes. There is no autocorrelation in the residuals, which does not make it normally distributed.

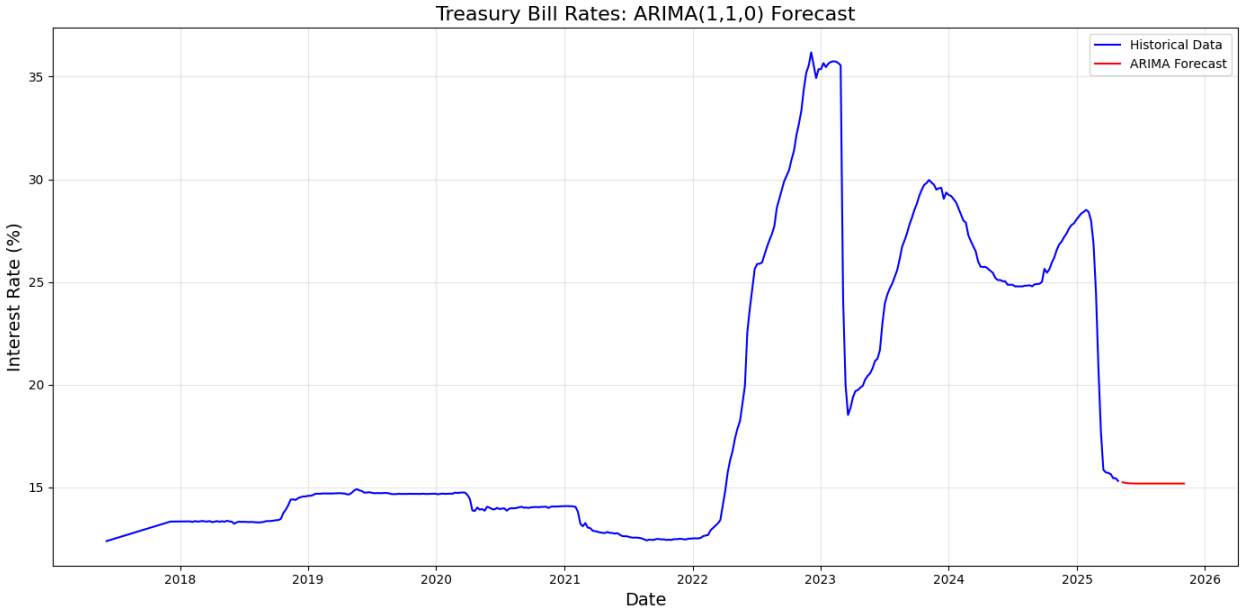


Figure 4.7 ARIMA (1, 1, 0) Forecast Figure 4 7 showing the ARIMA(1,1,0) forecast

Interpretation

The above model ARIMA (1, 1, 0) shows that the treasury bill rate of 91-day remained stable around 12 to 15% from 2018 through early 2022, and the rate began a dramatic upward trend at approximately 36% in 2023. After this rise, the rates faced significant volatility with multiple increases and decreases. In 2025, the rates dropped sharply from around 28% to about 15%. The ARIMA (1, 1, 0) model forecasts that the interest rates will remain stable at approximately 15% into 2026.

4.4 SARIMA Model

Table 4.4 Table 4 4 showing the SARIMA results

4.4.1 SARIMA (1, 1, 1) (1, 1, 0, 52)

Term	Values
Log Likelihood	-392.92
AIC	793.84

BIC	809.01
HQIC	799.89

Interpretation

From the above table, it is evident that the model has both non-seasonal and seasonal components, where the non-seasonal AR (1), first-differencing, and MA (1) components capture short-term patterns and trends. Also, the seasonal AR (1) with seasonal differencing at lag 52, which suggests weekly data with seasonal patterns.

Statistical Insights

The seasonal differencing component suggests that the data has a strong seasonal trend that needs to be removed. The inclusion of both seasonal and non-seasonal AR terms indicates firm in both short-term and seasonal patterns.

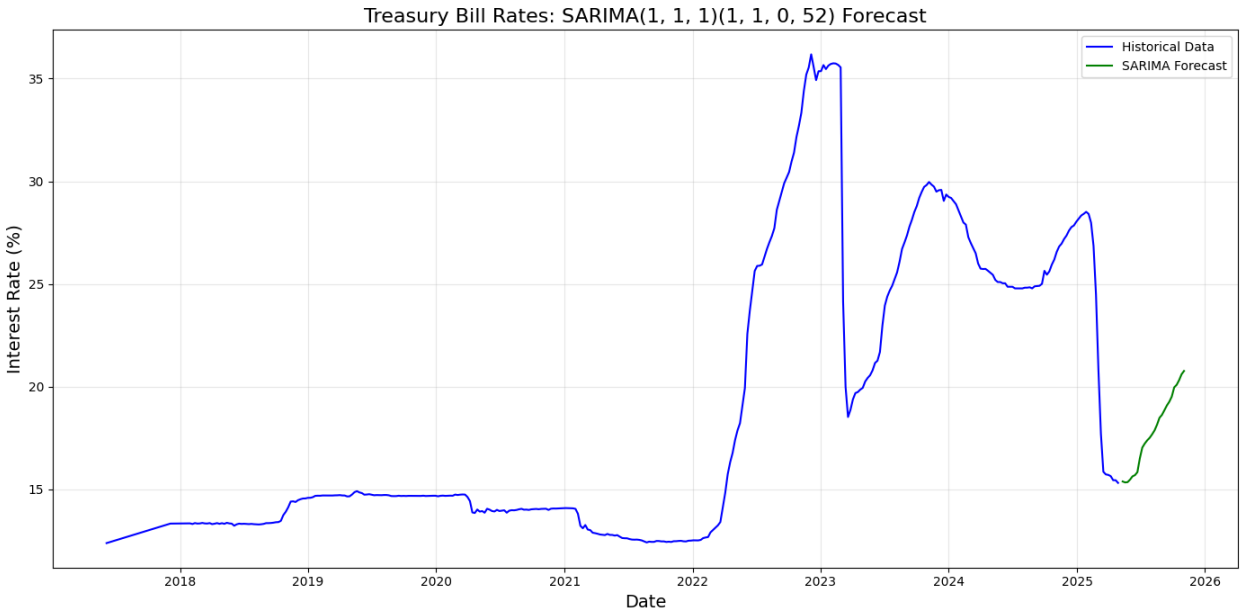


Figure 4.8Figure 4 8 showing the SARIMA graph

Interpretation

The historical data in blue tracks the interest rate over time, indicating that things were relatively stable, hovering around 12% to 15% from late 2017 to mid-2022. Then, a sudden change occurred in the late first quarter of 2022 to 2023, fluctuating in the 20s and 30s percent, respectively, in 2024. Then in 2025, there's a dip again in it early, and a slight increase. The SARIMA model is predicting a modest increase from 15% to 20% by early mid-2026.

4.4.1 SARIMA Residual Diagnostic

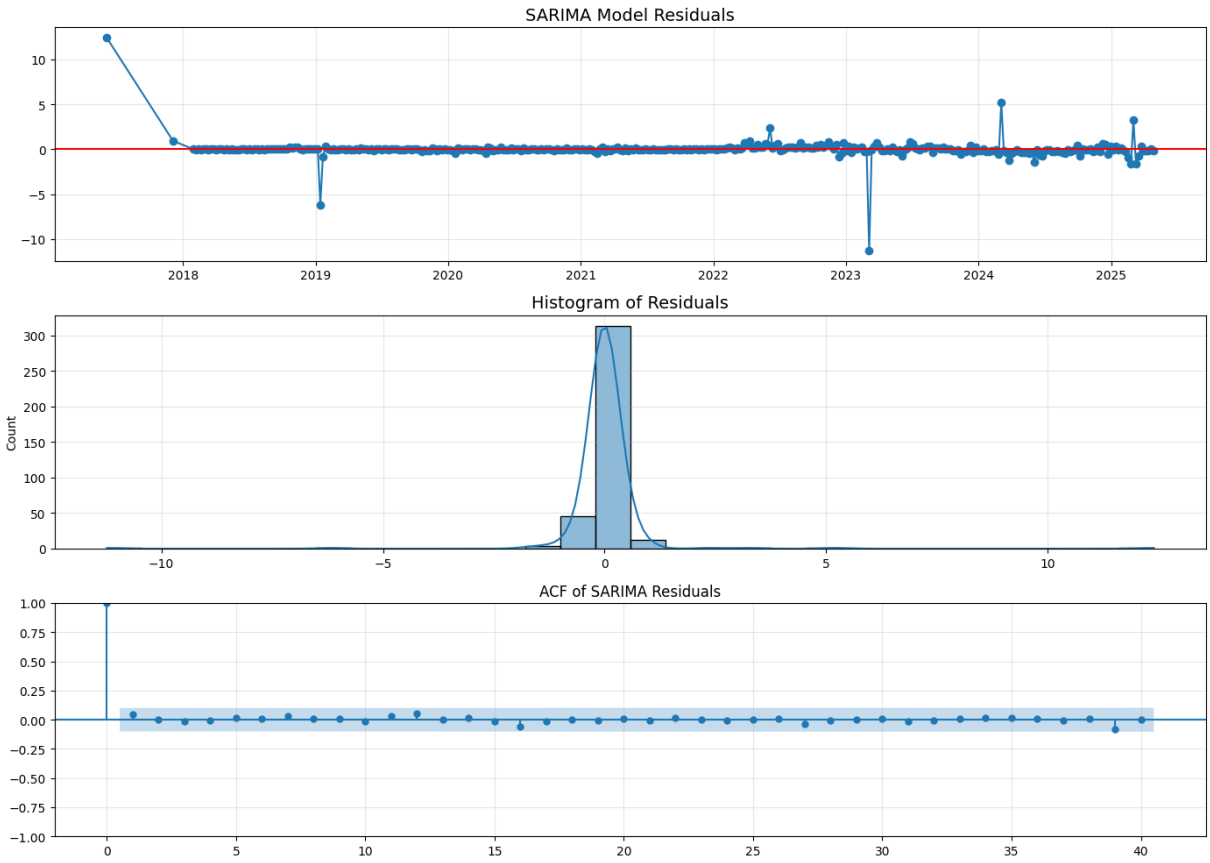


Figure 4 9 showing the SARIMA residual diagnostic

Interpretation

From the above graphs, the first plot indicates that the model has captured both the trend and seasonality well. Also, the second graph shows that most of the residuals are clustered near zero, which is good. Finally, the third graph indicates that there's no autocorrelation in the residuals.

4.5 Holt-Winters Multiplicative Forecast (Exponential Smoothing).

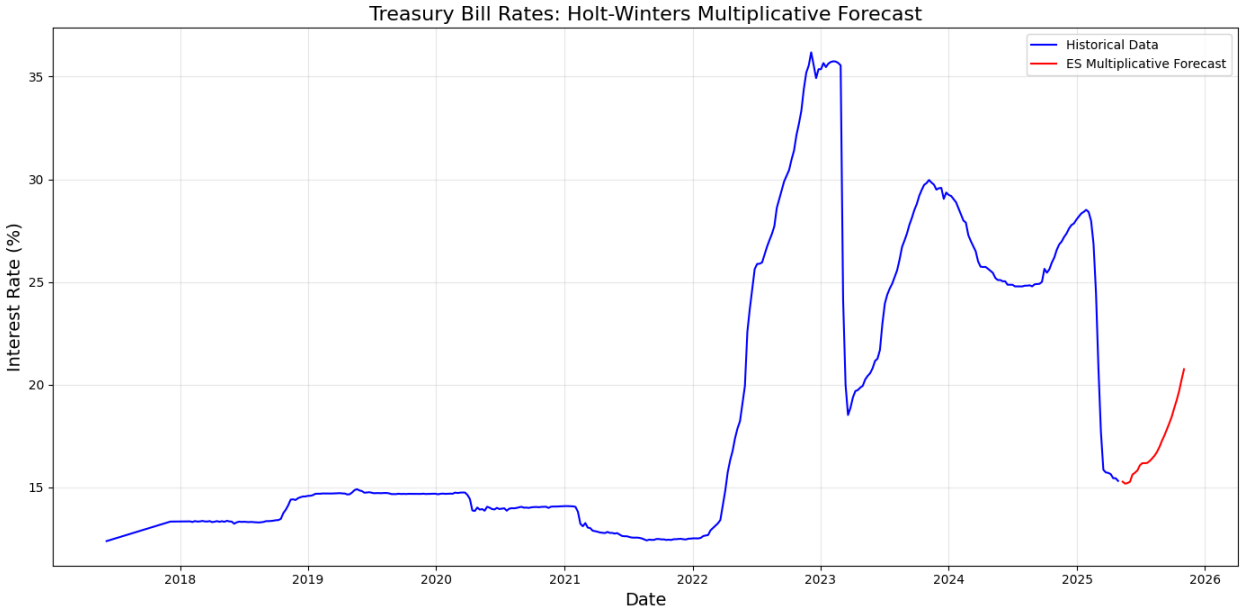


Figure 4 10 showing the Holt-Winters Multiplicative Forecast

Interpretation

The Holt-Winters multiplicative forecast effectively captures the seasonal and trend components of Ghana's 91-day Treasury bill rate and predicts a substantial increase in the rate after a notable historical move. However, the model's sensitivity to recent data spikes suggests that it warrants careful interpretation and validation using comparative forecasting methods.

4.5 FACEBOOK PROPHETS

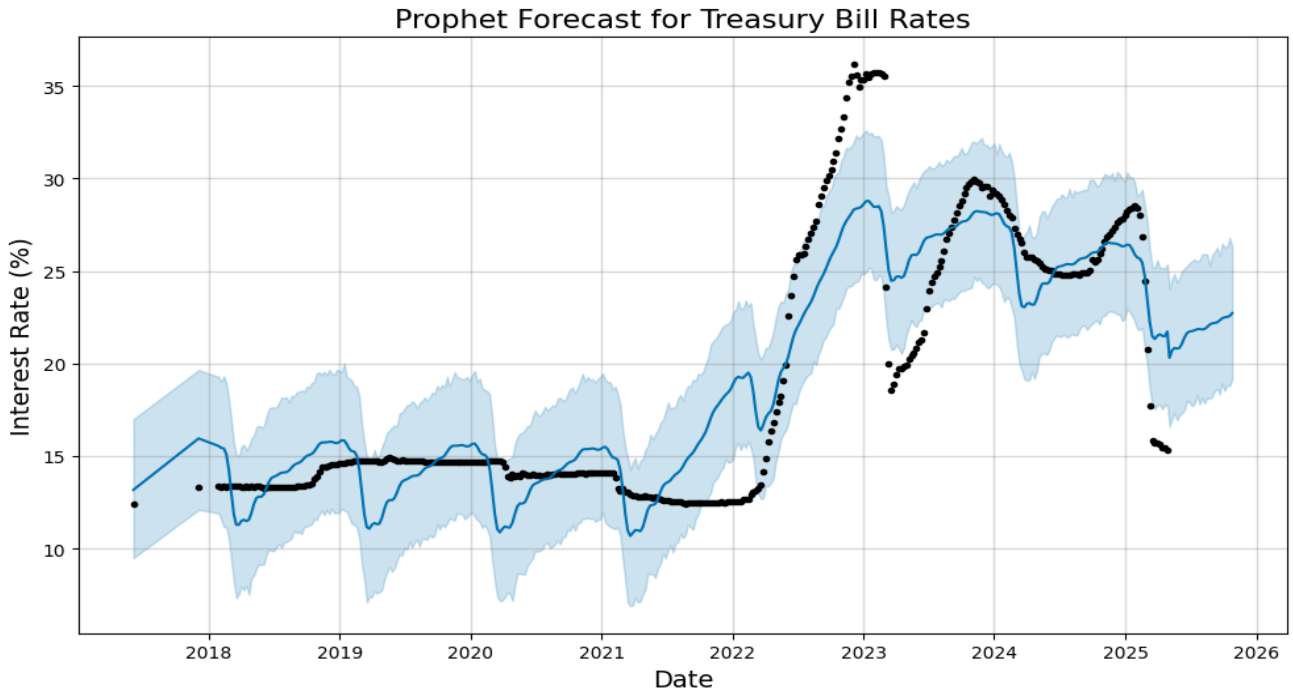


Figure 4 11 showing the prophet forecast for Treasury Bill Rates

Interpretation

The prophet forecasting model captured both trend and seasonal patterns in the data. At a wide confidence interval, which suggest the model recognized that the current environmental change occurred from 2022 to 2023.

4.6 ARTIFICIAL NEURAL NETWORK

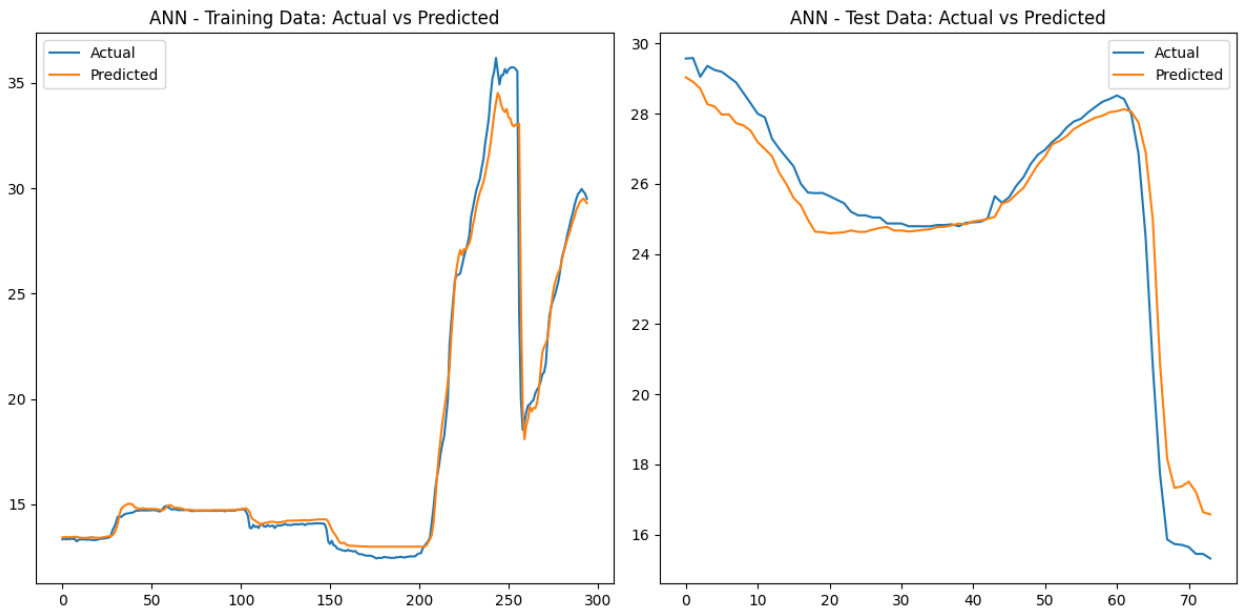


Figure 4 12 showing the results for graphs of Artificial Neural Network

Interpretation

From the above graphs, the ANN model exhibits excellent performance on training data and strong generalization on test data.

4.6 LONG-SHORT TERM MEMORY

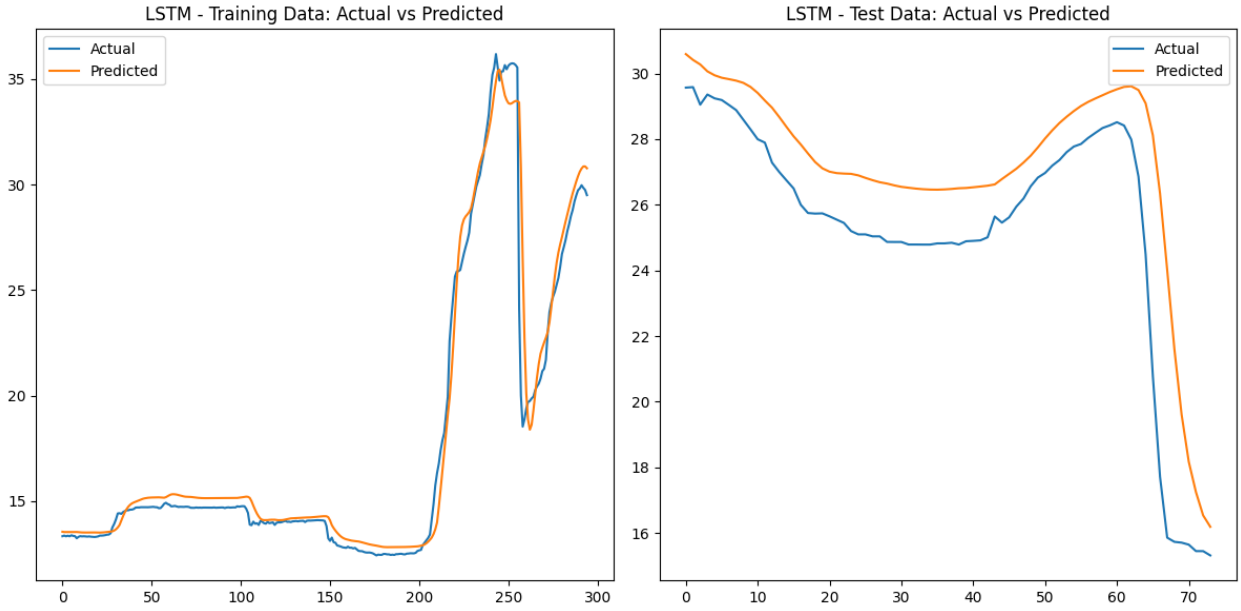


Figure 4 13 showing the Long-Short Term Memory

Interpretation

From the above graph, the LSTM model performs reasonably well on training data but suffers from overprediction and lag on test data.

4.7 CONCLUSION

This chapter has provided a comprehensive analysis of Ghana's 91-day Treasury Bill rate forecasting using multiple time series modeling approaches. This investigation, spanning from June 2017 to May 2025, with 381 daily observations, has yielded significant insights into the behavior and predictability of this critical monetary policy instrument.

4.7.1 Statistical Findings

The descriptive analysis revealed that Ghana's 91-day Treasury Bill rates showed a substantial variability over the study period, with a mean rate of 18.98% and a standard deviation of 6.97. The distribution demonstrated right-skewness, which indicates periods of exceptionally high rates that extends beyond the

typical range. This variability reflects the dynamic nature of Ghana's monetary environment and the influence of various macroeconomic factors on short-term interest rates. The time series decomposition analysis uncovered different phases in the Treasury Bill rate trajectory. The historical data exhibited relative stability between 2017 and mid-2022, with rates consistently maintained between 12% and 15%, indicating effective monetary policy management and economic stability. The dramatic structural break occurred in mid-late 2022, with the rates experiencing unprecedented volatility and reaching peaks of approximately 36% in 2023, before declining to stabilize around 15% in 2025.

4.7.2 Model Performance and Validation

The stationarity analysis confirmed the non-stationary nature of the original series, which called for first-order differencing to achieve stationarity. The Augmented Dickey-Fuller test results provided strong statistical evidence with a p-value equal to 0 for stationarity after differencing, validating the appropriateness of the ARIMA model. Among the traditional econometric models, the ARIMA (1,1,0) model showed superior performance with an AIC of 761.95, effectively capturing the autoregressive nature of the series while maintaining thrift. The model residual diagnostic displayed no significant autocorrelation, suggesting adequate model specification. The SARIMA (1,1,1) (1,1,0,52) model, while incorporating seasonal components, shows a higher information criterion (AIC = 793.84), also indicating that weekly seasonality wasn't a dominant feature in this daily Treasury Bill rate series. The Holt-Winters Multiplicative method effectively captured both trend and seasonal components, which demonstrates how the model's capability adapts to the dramatic structural changes observed in 2022 – 23.

The machine learning approaches (Artificial Neural Network, Facebook Prophet, and Long Short-Term Memory) yielded mixed results. The Facebook Prophet model successfully identified the structural break, which provided a wide confidence interval reflecting the increased uncertainty following the 2022 – 23 period. The ANN showed excellent training performance with strong generalization capabilities on the test data, and suggested robust pattern recognition. Also, the LSTM model performed adequately on the training data and exhibited overprediction tendencies and lag on test data, which indicated potential overfitting to historical patterns.

4.7.3 Forecasting Implications

Projections from various methodologies reveal several important implications for Ghana's monetary policy environment. First, all models indicate that the policy rate will stabilize in the range of 15–20% by 2026, signaling a return to more normal monetary conditions following the unusual volatility in 2022–2023. Despite varying forecast values, this convergence reinforces confidence in the precise direction of the forecast. The sharp increase in interest rates in 2022–2023, peaking above 35%, could reflect specific

economic circumstances, possibly related to inflationary pressures, monetary instability, or global economic turmoil. The subsequent decline and expected stabilization suggest that these conditions represent a temporary rather than a permanent change in Ghana's interest rate environment.

4.7.4 Research Objective Fulfilment

This analysis successfully achieved the primary objective of the study: to develop robust forecasting models for the 91-day Ghana Treasury Bill interest rate. The comparative approach showed that traditional econometric methods (in particular ARIMA models) remain competitive with advanced machine learning methods for this specific application. The study concluded that the trade-off between model complexity and interpretability should be considered when choosing a forecasting method, as simpler models generally provide more robust and reliable forecasts for policymaking. The study has practical implications for a variety of stakeholders. Investors can use these forecasts for portfolio optimization and risk management, policymakers can better understand changing monetary conditions, and financial institutions can improve their asset and debt management strategies. The wide confidence intervals obtained from multiple models adequately capture the uncertainty inherent in monetary forecasts, especially following periods of structural change.

4.7.5 Limitations and Future Research Directions

While this analysis provides valuable insights into the dynamics of government bond yields, it has several limitations. The structural breakdown in 2022–2023 poses a challenge for all forecasting models, as historical records may not fully reflect the unprecedented economic conditions. The relatively short time series limits the assessment of model performance across multiple business cycles. Future research should consider macroeconomic variables such as inflation rates, exchange rates, and global interest rate trends to improve forecast accuracy. Furthermore, combined forecasting methods that combine forecasts from multiple models can produce more robust forecasts than individual methods. The results contribute to the growing literature on interest rate forecasting in emerging markets and demonstrate the continued validity of traditional time series methods in modern monetary analysis. The study's integrated approach, which combines multiple forecasting paradigms, provides a solid framework for understanding the complex dynamics of short-term interest rates in Ghana and delivers practical tools for financial decision-making in this important West African economy.

CHAPTER FIVE

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

5.1 INTRODUCTION

This final chapter provides a comprehensive summary of the study "Time Series Forecasting Using 91-Day Treasury Interest Rates in Ghana". The study uses various forecasting techniques to analyze and forecast the Treasury interest rates in Ghana from June 2017 to May 2025. Both traditional econometric methods and modern machine learning techniques are used. This chapter summarizes the main findings, discusses the limitations identified during the study, draws conclusions on the performance of the various forecasting models, and provides practical recommendations for stakeholders in the financial sector in Ghana.

The study addressed key questions on the predictability of short-term interest rates in Ghana, the comparative performance of different forecasting methods, and the practical implications for monetary policy and monetary decision making. Through a systematic analysis using ARIMA, SARIMA, Holt-Winters exponential smoothing, Facebook Prophet, Artificial Neural Networks, and Long Short-Term Memory, this study provides valuable insights into the emerging market interest rate forecasting literature and offers practical tools for financial practitioners and policymakers.

5.2 LIMITATION

Despite the comprehensive nature of this study, there are several limitations that must be acknowledged that can affect the generalizability and applicability of the findings.

5.2.1 Data Limitations

This research relied exclusively on historical data without incorporating potentially influential macroeconomic variables such as inflation, exchange rate, fluctuations, and GDP growth. This univariate approach, while methodologically sound for the initial analysis, may have limited the models' ability to capture the full complexity of factors influencing interest rates in Ghana. Furthermore, the relatively short time series, covering about eight years, may not adequately capture long-term cyclical trends or provide sufficient data for reliable testing across multiple business cycles.

5.2.2 Structure Break Challenges

The dramatic structural volatility of interest rates, from approximately 15% to over 36% in the 2022-2023 period, poses a significant challenge for all forecasting models. This period of

unprecedented volatility could represent unique economic conditions that are not adequately captured in historical data and therefore cannot be adequately predicted by models. The unique nature of this period raises questions about the ability of models to predict similar volatilities in the future.

5.2.3 Model Limitations

Each forecasting model used in this study had inherent limitations. Traditional time series models (ARIMA, SARIMA) assume linear relationships and can have difficulty dealing with nonlinear patterns in financial data. Machine learning methods are more flexible but often require large datasets for optimal performance, which can lead to overfitting, as seen with the LSTM model. Exponential smoothing methods show high sensitivity to recent extreme values, which can make forecasts less reliable during periods of high volatility.

5.2.4 External Validation Constraints

The validation of this study is limited to segments of historical data and lacks long-term, real-time, out-of-sample testing. This makes it difficult to assess the model's performance in real-world deployment scenarios where new and unprecedented economic conditions may arise.

5.3 SUMMARY

This study systematically examines the effectiveness of various time series methods in predicting the 91-day interest rate on Ghana's treasury bills. The study period shows different phases: a stable period from 2017 to early 2022, with interest rates between 13% and 15%, abnormal fluctuations in 2022 and 2023, with peaks exceeding 36%, and then a stabilization of around 15% until early 2025.

5.3.1 Key Findings

The descriptive analysis shows that the Treasury bond yields exhibit a significantly right-skewed distribution with a mean of 18.98% and a standard deviation of 6.97%, indicating considerable fluctuations during the sample period. Stationarity tests confirm that first-order differences are necessary to achieve statistical stationarity. The results of the extended Dickey-Fuller test clearly distinguish between the original nonstationary series and the stationary difference series. Among the traditional econometric models, the ARIMA(1,1,0) model had the lowest AIC value of 761.95, and showed better performance by effectively capturing the autoregressive characteristics of the data after appropriate difference processing. The SARIMA model incorporated seasonal factors, but the forecast accuracy did not improve significantly. The Holt-Winters multiplicative exponential smoothing method coped well with trend and seasonal fluctuations but showed sensitivity to extreme values during the volatile 2022–2023 period. Advanced

machine learning models produced mixed results. Facebook Prophet correctly detected structural changes and provided realistic confidence intervals. Artificial Neural Networks demonstrated good generalization capabilities, but the LSTM model exhibited overprediction issues on the test data.

5.3.2 Convergent Forecasting Results

Despite the methodological challenges, most of the models converged on forecasting the treasury bill rate to stabilize around a 15 to 20 % range through 2026, and suggesting a return to normal following a period of extraordinary volatility, financial conditions have normalized. Convergence between different modeling methods provides greater confidence in the directional accuracy of the forecasts.

5.4 CONCLUSION

This study has successfully demonstrated a multiple time series forecasting method that can provide valuable insights into Ghana's treasury bill rate dynamics, each providing a unique strength and limitations. The comparative study approach shows that the traditional econometric methods, particularly the ARIMA (1,1,0) model, remain highly competitive with the other models. The big structural rupture that happened in 2022–2023 showed how hard and how easy it is to make financial predictions. This period presented challenges to all the models, but also highlighted the importance of using a variety of forecasting methods and maintaining appropriate uncertainty measures in forecasts. The stability and forecast convergence results for the various methods since then suggest that the Ghana interest rate volatility reflects temporary fluctuations rather than permanent changes.

This study contributes to the growing literature on market finance by highlighting the importance of machine learning methods and demonstrating the robustness of traditional time series methods. The results support the view that simpler and more well-defined models tend to produce more accurate and understandable forecasts for policy applications than more complex models, especially when dealing with financial time series with structural breaks. From a pragmatic perspective, this study provides a useful resource for various participants in the Ghanaian financial system. Investment choices of financial institutions, monetary policy formulation, and risk management procedures can be improved by using forecasting models. Since financial forecasts are inherently uncertain, wide confidence intervals can help in decision making.

5.5 RECOMMENDATION

Based on the findings and analysis, the following recommendations are proposed for different stakeholders.

5.5.1 Policymakers

Monetary policy authorities should consider incorporating multiple time series methodologies into their analytical framework, rather than relying solely on single-model approaches. The study results suggest that the 15-20% range should inform medium-term policy planning, while the wide confidence intervals should exhibit flexible policy responses to unexpected development. The central bank should begin investing in the development of macroeconomic forecasting models that incorporate treasury bill rate predictions alongside other key economic indicators. The structural break observed in 2022–2023 emphasizes the importance of maintaining robust early warning systems for detecting emerging financial stress.

5.5.2 Future Research

Future research should expand the analysis to incorporate multivariate models that include macroeconomic variables such as inflation, exchange rates, and global trends. This would improve the accuracy and provide deeper insights into the drivers of the treasury bill rate mechanism.

REFERENCES

Acquah, A., & Khormortey, E. (2019). Treasury bills as financial instruments: A theoretical analysis. *Journal of West African Finance*.

Acquah-Sam, E., & Salami, K. (2014). Effect of selected macroeconomic variables on treasury bill rates in Ghana. *Journal of Economics and Sustainable Development, 5*(4), 46-55.

Adam, A. M. (2014). Modelling and forecasting interest rates in Ghana: An application of ARIMA models. *International Journal of Economics and Finance, 6*(8), 121-128.

Addison, T. (2018). Treasury bill rate forecasting in developing economies: Evidence from Ghana. *African Development review, 30*(2), 145-162.

Addo, E., Mensah , K., & Asante, P. (2020). Treasury bill auction mechanics in Ghana: Analysis of pricing and market dynamics. *Ghana Economic Review*, 89-107.

Agyekum, F. (2021). Digital transformation and financial market efficiency in Ghana. *African Journal of Finance and Technology*.

Akosah, P., Mensah , K., & Asante, J. (2022). Machine learning approaches for interest rate forecasting in Ghana: A comparative analysis. *Journal of African Financial Technology*, 78-95.

Alagidede, P., & Mensah, J. (2018). Financial sector reforms and market stability in Ghana. *Journal of Banking and Finance in Africa*.

Anoki, K., & Toyneboah, R. (2009). Nonlinear dynamics in Ghanaian treasury bill rates. *West African Journal of Monetary and Economic Integration, 9*(1), 23-45.

Antwi, S., Mensah , K., & Boateng , A. (2020). The evolving role of treasury bills in Ghana's financial architecture. *Ghana Economic Review*, 123-140.

Ayitey, J. Z. (2019). Data quality and financial market analysis in Ghana. *Journal of Developing Country Finance*, 78-92.

Beckbe, M. (2011). Data limitations in African financial markets: Challenges for econometric analysis. *African Economic Research Consortium Working Paper*, 234.

Biekpe, N., & Adjasi, C. (2006). Stock market development and economic growth: The case of selected African countries. *Global Economy Journal*, 1-21.

Boadi, L., & Amegbe, H. (2017). Liquidity preference and treasury bill pricing in Ghana's financial market. *Journal of African Finance*, 145-162.

Boateng, E., & Amponsah, S. (2020). Retail investor participation in Ghana's treasury bill market: Drivers and implications. *African Journal of Business Management*, 234-251.

Bomia, M., & Abrado-Otto, K. (2003). Monetary policy transmission mechanisms in Ghana. *Bank of Ghana Working Paper*.

Bondzi-Simpson, A., Aryetey , F., & Danquah , L. (2019). Evolution of Ghana's financial market: Three decades of transformation. *West African Economic Journal*.

Dzigbede, K., & Ofori, I. (2022). Expectations theory and Ghana's yield curve: Evidence from 2017-2021. *African Finance and Economics Review*, 189-206.

Frimpong , J. M., & Oteng-Abayai, E. F. (2008). Modelling and forecasting volatility of returns on the Ghana Stock Exchange using GARCH models. *American Journal of Applied Sciences, 5*(4), 389-396.

Kumi, E. (2023). Institutional theory of interest rates in developing economies: Evidence from West Africa. *Journal of Development Economics*.

Kwakye, J. K. (2021). Classical loanable funds theory and interest rate determination in Ghana. *Bank of Ghana Journal*, 12-28.

Kwakye, J. K. (2022). Fiscal dominance and interest rate dynamics in Ghana. *African Economic Policy Review*, 89-107.

Mensah, S. (2013). Interest rate volatility and banking sector performance in Ghana. *African Journal of Economic and Management Studies, 4*(2), 167-181.

Mensah, S. (2015). Forecasting uncertainty in Ghanaian financial markets. *Journal of Risk and Financial Management, 8*(2), 234-251.

Oduro-Afriyie, K., & Adeyemi, B. (2020). Public debt dynamics and fiscal sustainability in Ghana. *Journal of African Public Affairs, 13*(1), 89-104.

Osei, K. A. (2002). Asset pricing and information efficiency of the Ghana Stock Market. *African Economic Research Consortium Research Paper*, 115.

Osei-Assibey, E., & Asenso, J. K. (2019). Investor base analysis of Ghana's government securities market. *West African Capital Markets Journal*, 189-207.

Quarcoo, J., & Allen, F. (2020). Institutional framework for government securities management in Ghana. *Public Financial Management in Africa*, 156-173.

Quartey, P., & Gameli, S. (2021). Foreign investor participation and domestic debt market dynamics in Ghana. *International Finance Review*, 145-162.

Quartey, P., Ackah, C., & Agyei-Holmes, A. (2017). Managing public debt in Ghana: Challenges and prospects. *African Development Review, 29*(4), 550-563.